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(54) **PIPE BURST PREVENTION RELIEF VALVE,
CONTROL VALVE SYSTEM AND METHOD**

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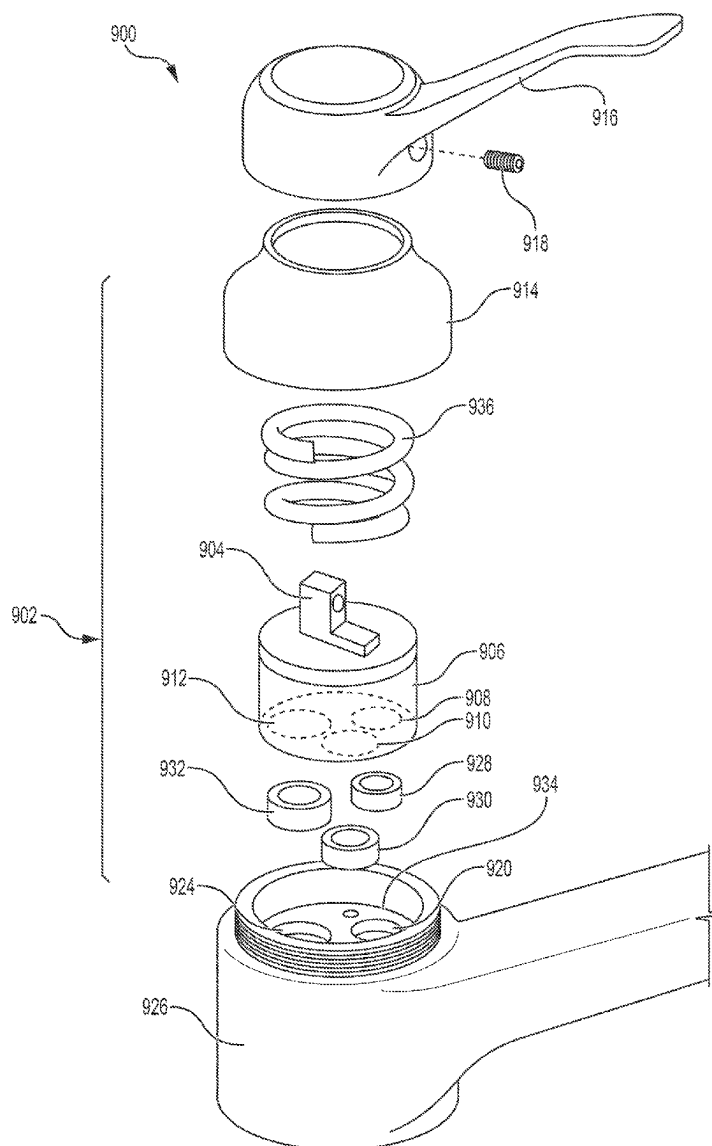
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ABSTRACT

A pressure relief valve that is spring-activated or utilizes a burst disk is described for preventing pipe bursts in a building plumbing system. A control valve system and method for preventing pipe bursts utilizing the pressure relief valve are also described. Also described are spring-activated control valve cartridges and control valve systems utilizing spring-activated control valve cartridges that prevent pipe bursts when installed building plumbing systems. Further described are methods for preventing pipe bursts in building plumbing systems comprising the step of disposing a relief valve or spring-activated cartridge according to the disclosure in a building plumbing system.



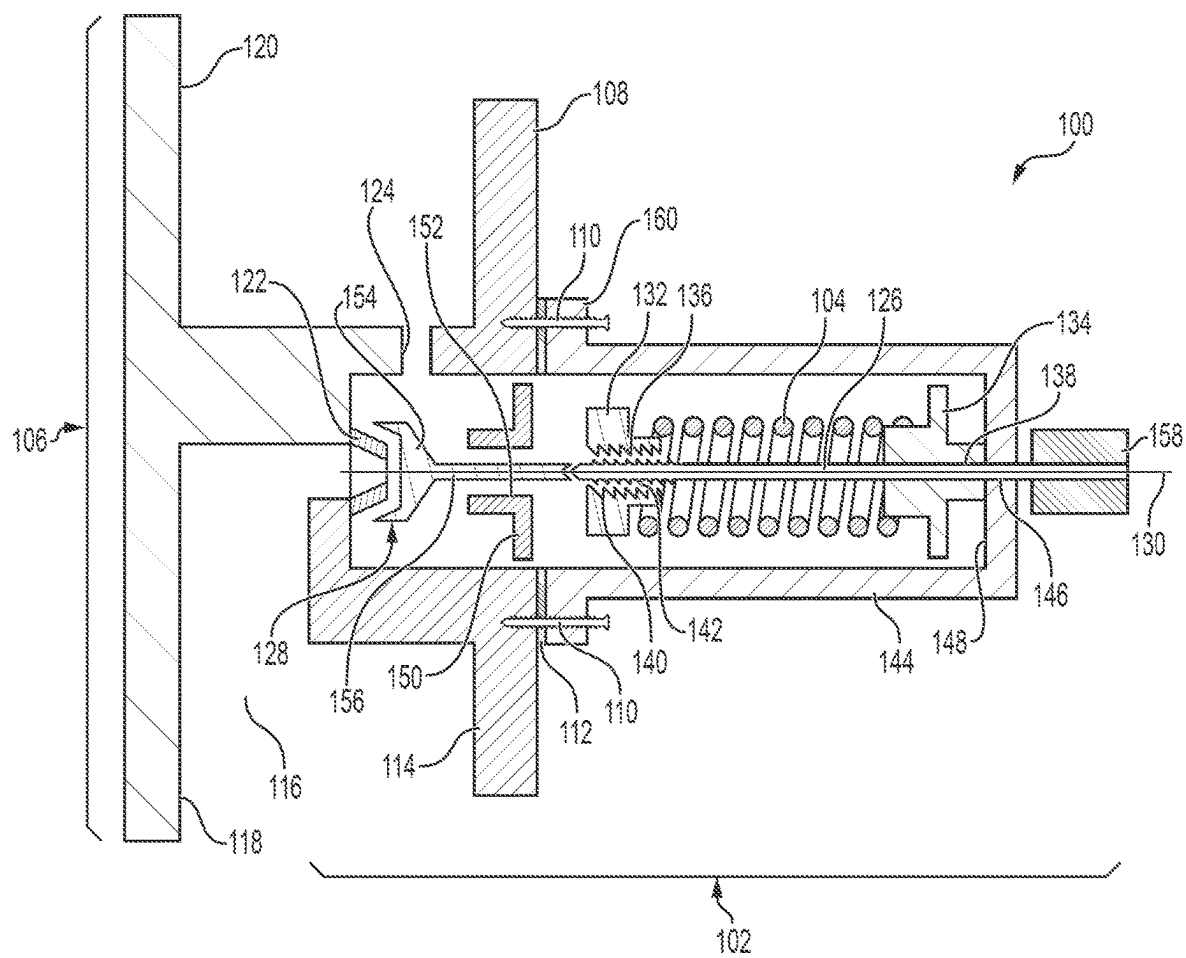


FIG. 1

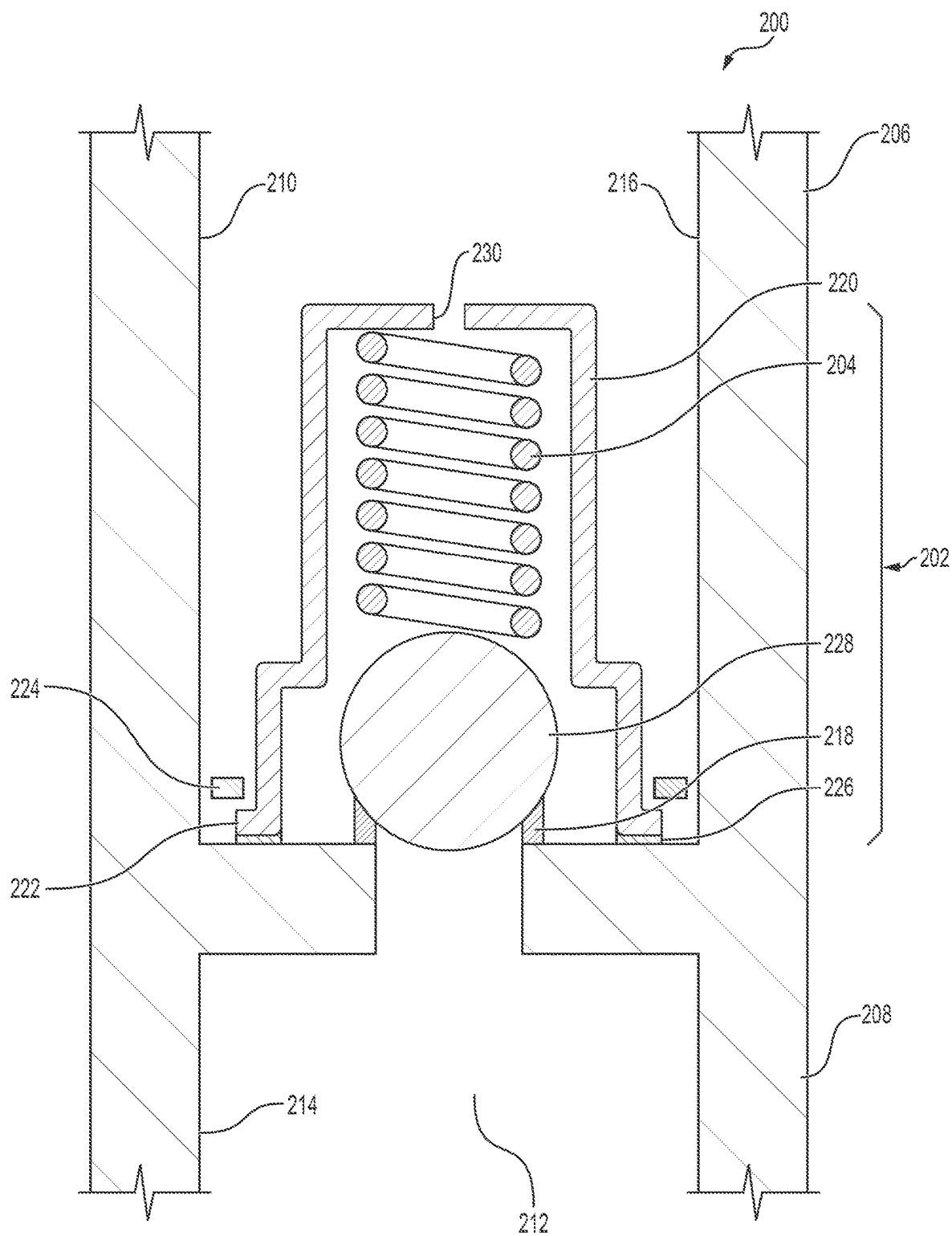


FIG. 2

FIG. 3

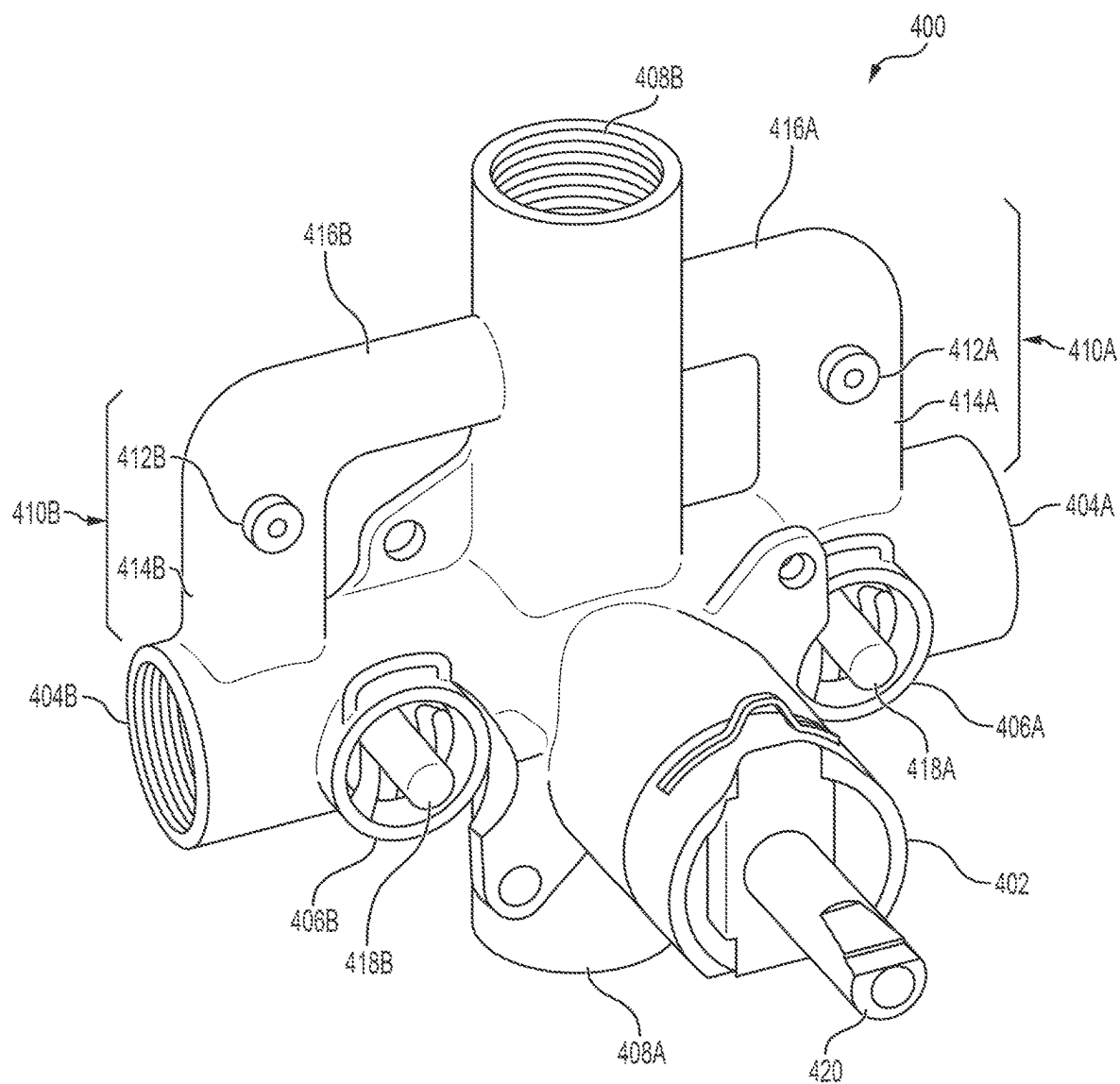


FIG. 4

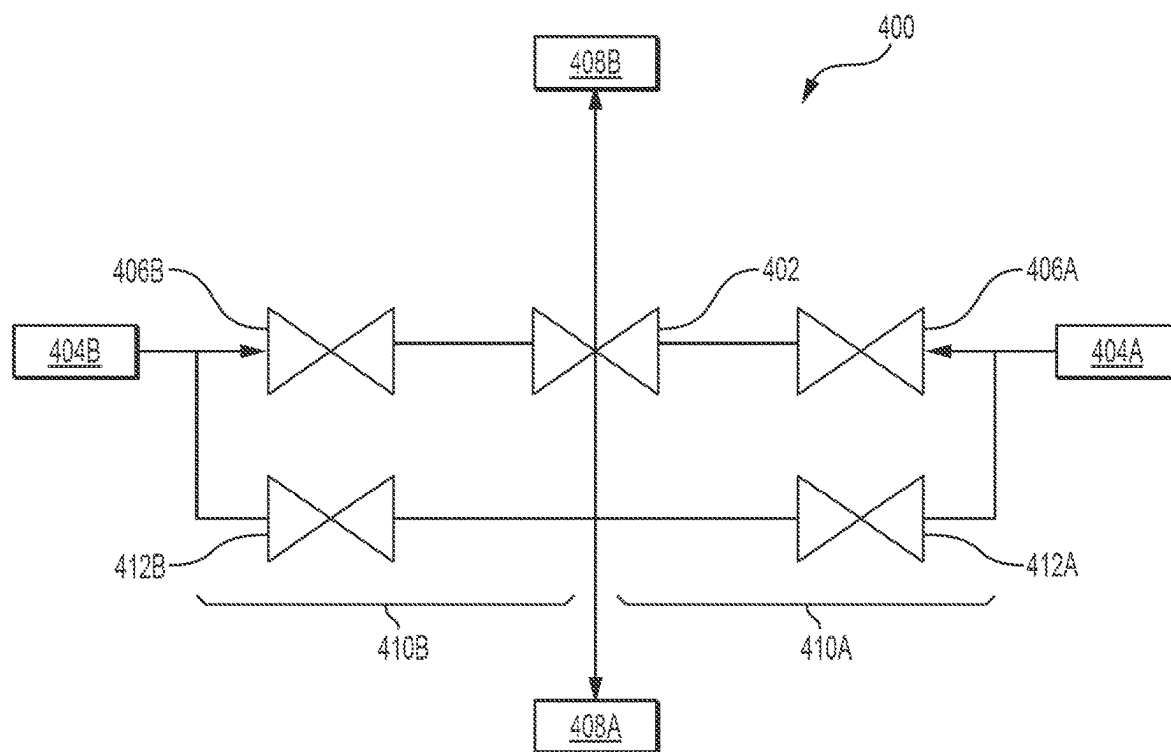
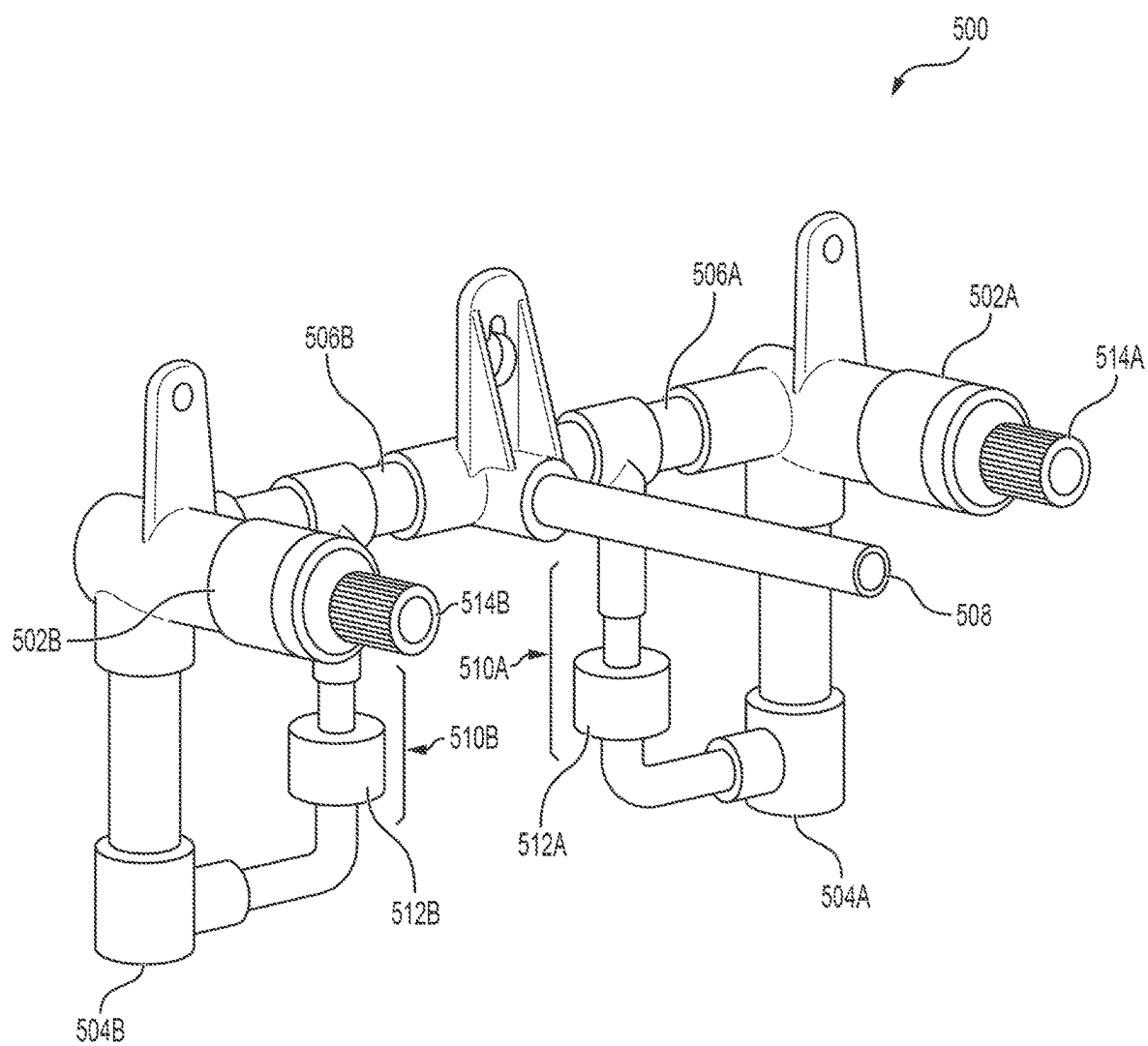


FIG. 5

**FIG. 6**

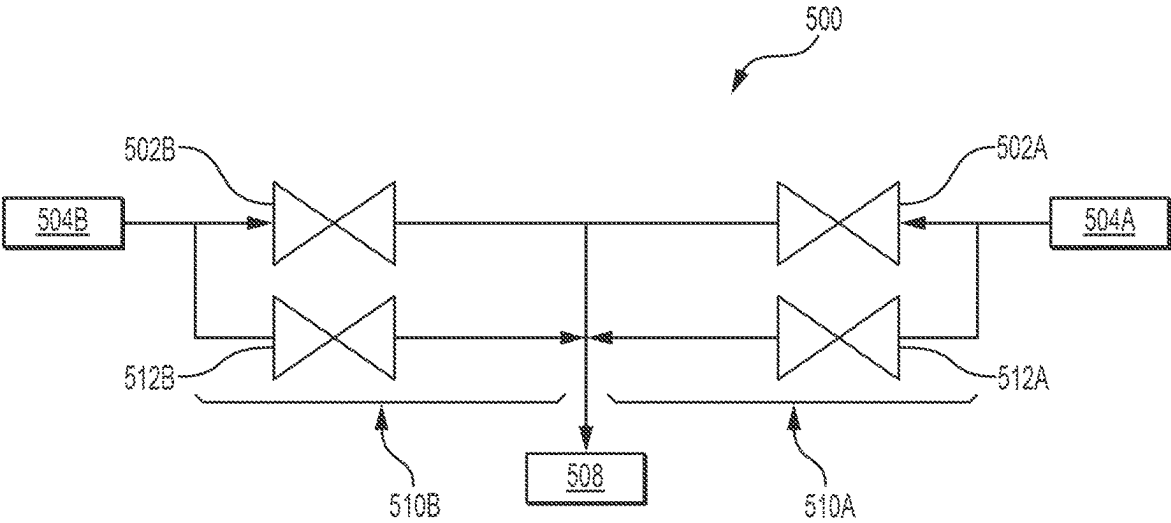


FIG. 7

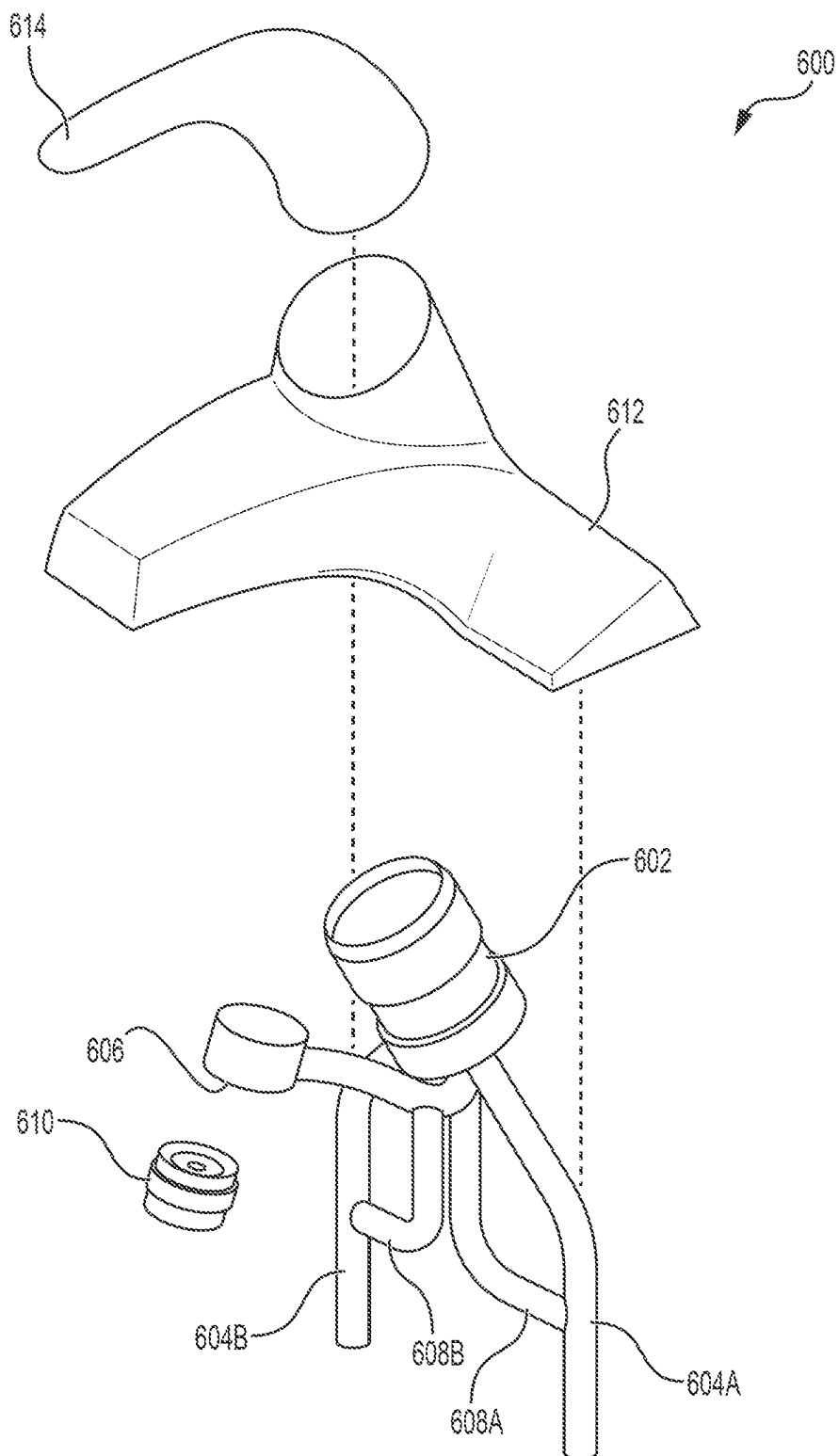


FIG. 8

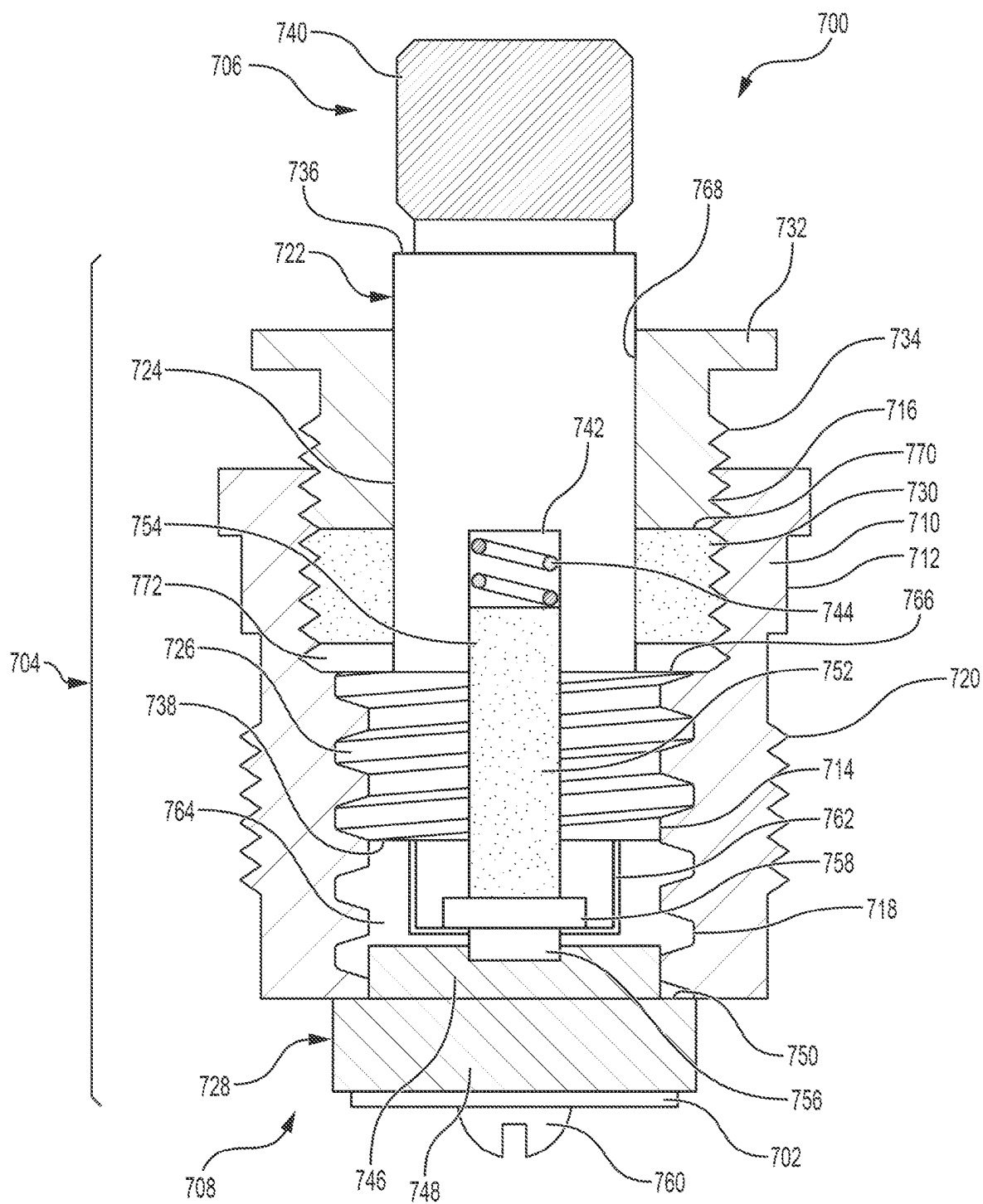


FIG. 9

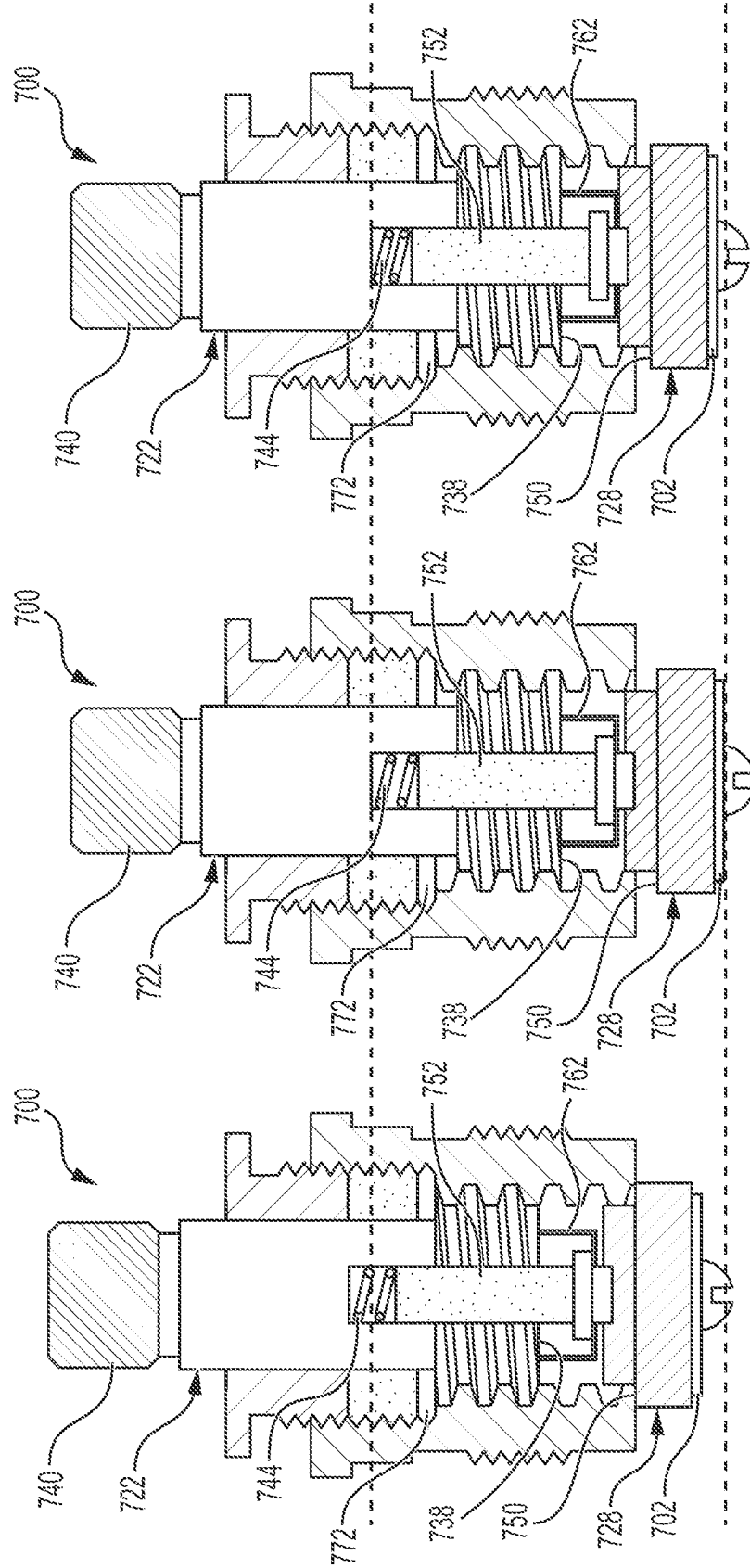
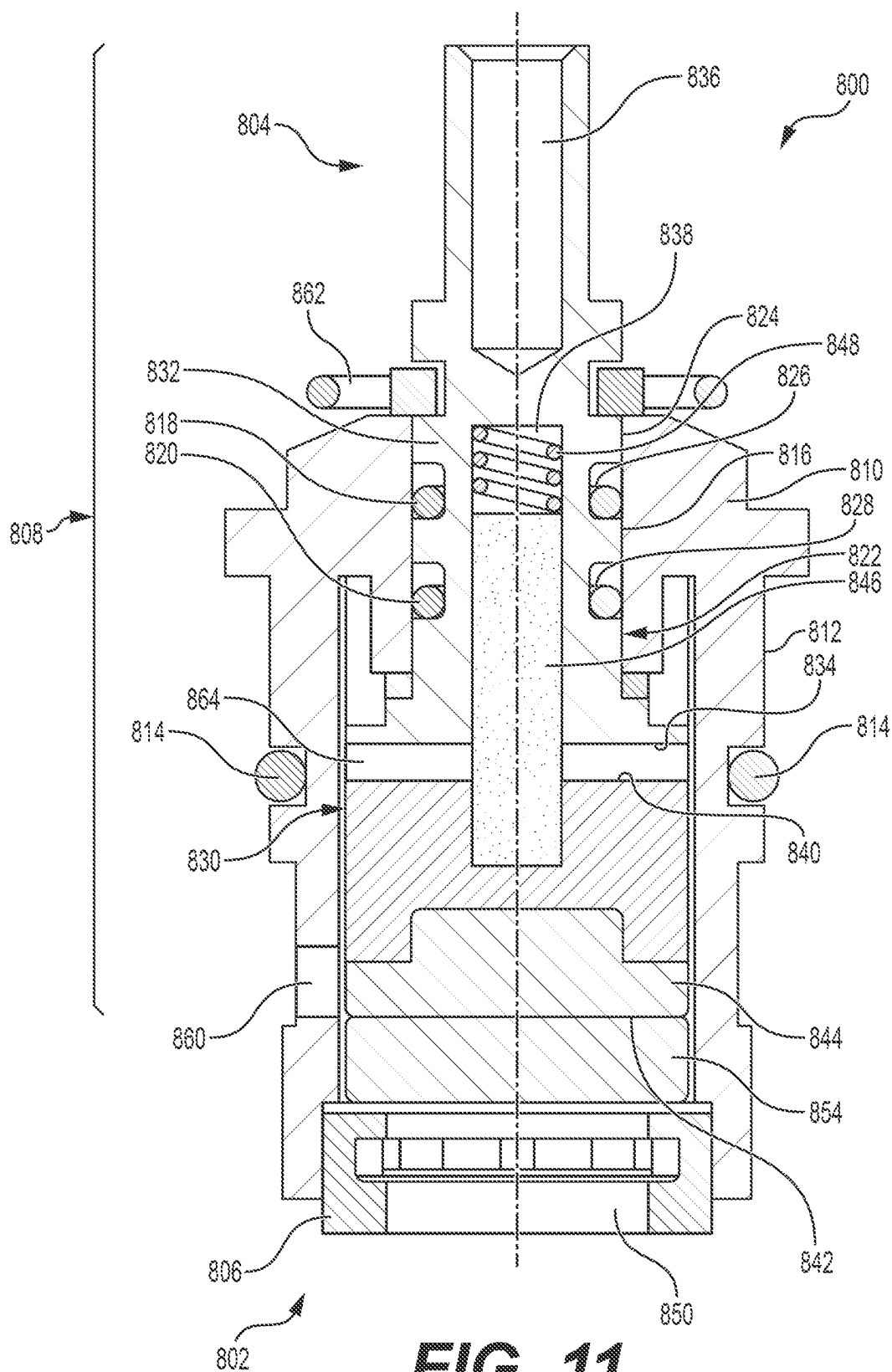


FIG. 10A **FIG. 10B** **FIG. 10C**



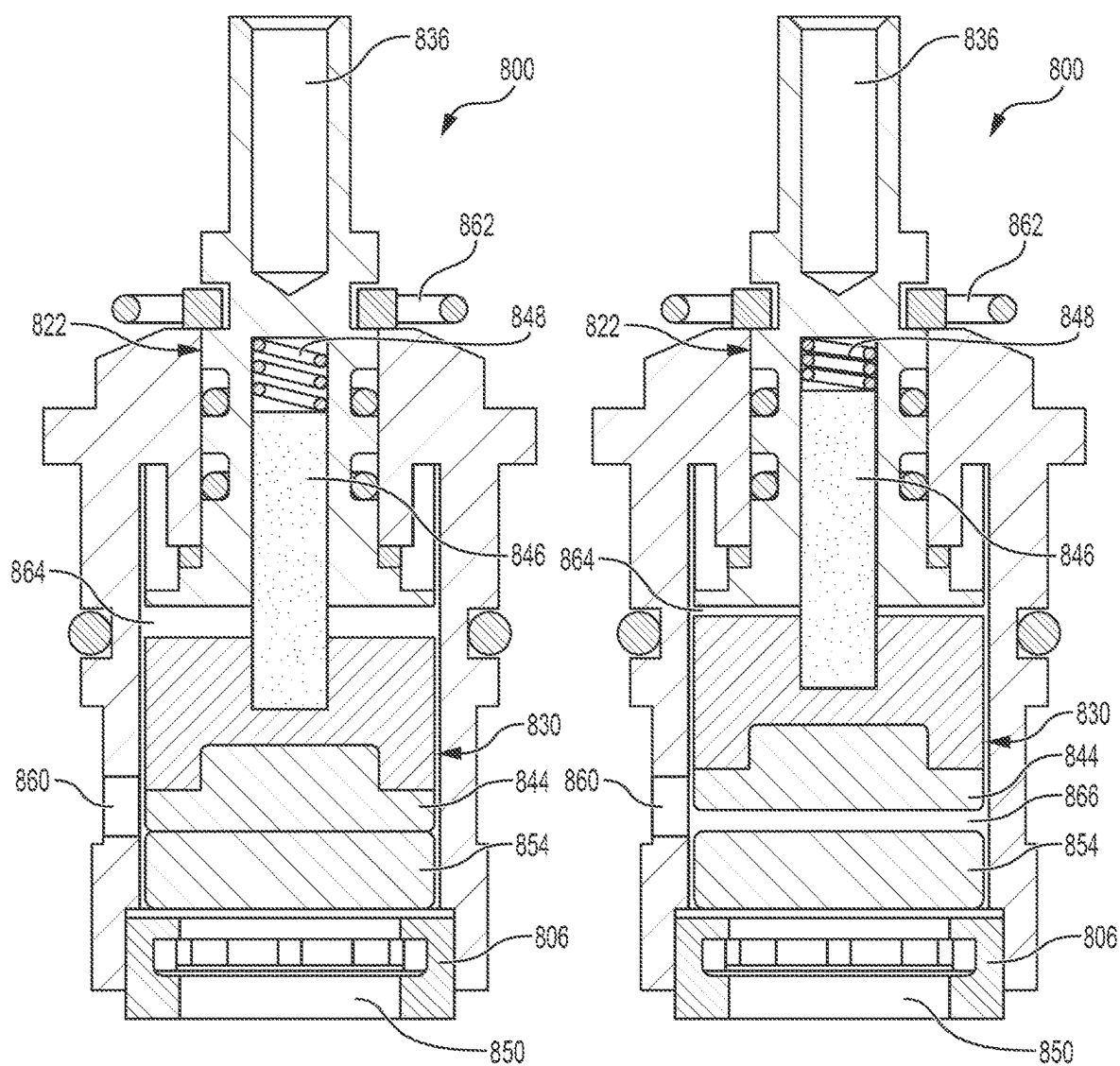


FIG. 12A

FIG. 12B

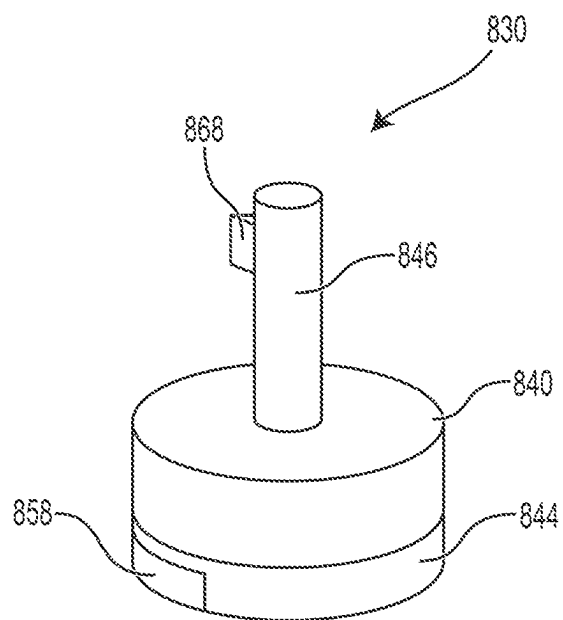
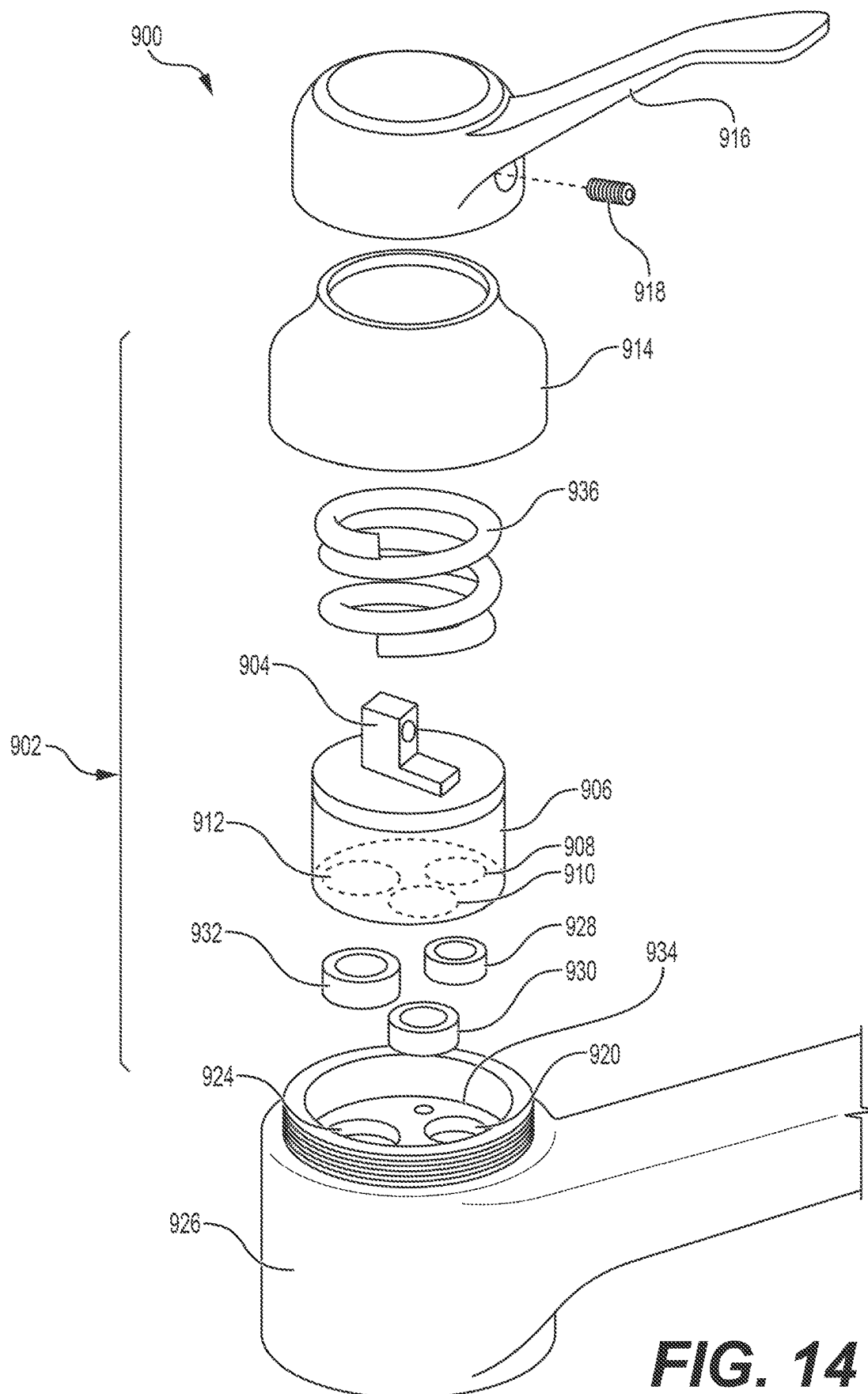


FIG. 13



PIPE BURST PREVENTION RELIEF VALVE, CONTROL VALVE SYSTEM AND METHOD

BACKGROUND

[0001] During cold snaps, water in pipes and valves in a building can freeze. When water in pipes and valves freezes, the pipes and valves can burst, damaging the plumbing, electrical and HVAC systems in the building and the area in the vicinity of the burst. Damage associated with pipe bursts can require that the water and other utilities to and systems in the building be turned off, and building occupants evacuate the building and the building remain unoccupied until it has been repaired.

[0002] Repairs associated with a burst pipe or valve can require the replacement of the pipe, valve, and wall in which the pipe or valve was installed, insulation, electrical and HVAC systems, fixtures and finishes in the surrounding building area, and furniture and other possessions in the building. Repairing the building and replacing damaged fixtures, furniture and possessions can cost a considerable amount of money. In addition, the inability to use the pipe or valve, and surrounding area can cause downtime, inconvenience and possibly additional costs associated with loss of use of the building, including rental income, and moving into and renting a temporary space until the repairs are completed.

[0003] It has been reported that burst pipes due to freezing cause over USD \$100 million in the United States and approximately USD \$1 billion worldwide in damages each year. The Red Cross has reported that more than 250,000 families across the United States experience frozen or burst pipes each year. In addition, losses due to burst pipes and valves are on the rise because of increases in the height of apartment buildings, the number of bathrooms per bedroom in dwellings, the number of water-using appliances in buildings, and the amount of water used per person. Losses due to burst pipes and valves are also rising because of the vulnerability of buildings to damage due to modern methods of building construction, such as tiny homes, modular buildings, bathrooms and kitchens, and mass timber or exposed beam construction.

[0004] Bursting pipes and valves in a building due to the freezing of water in them is a problem that persists to this day, despite efforts by others to solve the problem. What is needed, therefore, is a relief valve that can be installed in an existing building plumbing system which will protect it from bursting when pipes or valves in the system freeze. What is also needed is a compact water delivery control valve system for use as a plumbing fixture in a building plumbing system that does not burst and protects the pipes and other fixtures connected to it from bursting under freezing conditions. What is further needed is a method for preventing pipes or valves in a building plumbing system from freezing by disposing the relief valve or valve system having the relief valve in the building plumbing system.

SUMMARY

[0005] Accordingly, in one aspect, a relief valve for pipes or valves or other fixtures in a building plumbing system is described that includes a body having an inlet, an outlet, an exterior surface, and an interior surface that defines a generally longitudinal passageway that extends from the inlet to the outlet, a valve seat disposed on the interior

surface of said body between the inlet and the outlet, a flow limiter disposed on the interior surface of said body between said valve seat and the outlet, and extending into the passageway, and a valve seat sealing portion disposed in fluid communication with said valve seat between the inlet and said flow limiter, and configured to seal said valve seat against the flow of fluid in the passageway when the fluid has a pressure that is less than a predetermined pressure. In another aspect, a compact, pipe-burst-prevention valve system for use in a building plumbing system is described that includes a faucet housing having a handle and an outlet for delivering fluid to an end user, a control valve having an inlet and an outlet in fluid communication with each other and the control valve outlet in fluid communication with the faucet outlet, and the relief valve. Further described is a method for preventing pipes from bursting that includes disposing the pressure relief valve or valve system having the pressure relief valve in fluid communication with a building plumbing system.

[0006] Accordingly, in another aspect, a pipe-burst-prevention fluid control valve cartridge for fluid having a pressure is described that includes a body having a stem end, a valve end capable of being held in fluid communication with a fluid control valve of a faucet in a plumbing system, an exterior surface, and a generally cylindrical interior surface, a generally cylindrical split stem, at least a portion which is configured to be rotatably held within the interior of said cartridge body, having a spindle configured to receive a faucet handle, an upper stem sized so as to be received within the interior surface of said cartridge body and configured to be capable of being rotated when the spindle is turned, and an upper end adjacent to the spindle and a lower end generally opposite the upper end and adjacent to the cartridge body valve end with a cavity extending therefrom into the upper stem toward the upper stem upper end, the upper stem configured to extend toward the valve end of the cartridge body when the spindle is rotated to close the valve and retract from the valve end when the spindle is rotated to open the valve, a lower stem having an upper end configured so that a portion of it can be slidably received into the interior of said cartridge body at its valve end and adjacent to the lower end of the upper stem with a piston affixed thereto, and a spring held in the cavity by the piston that is capable of compressing when the pressure of the fluid in the valve equals or exceeds a predetermined pressure, wherein the piston, cavity and spring have diameters such that the cavity can receive slidably the spring and the piston and lengths such that the lower end of the upper stem and the upper end of the lower stem form a gap when the spring is fully extended. In another aspect, a compact, pipe-burst-prevention control valve system for use with fluid having a pressure in a building plumbing system is described that includes a faucet housing having a handle and an outlet for delivering fluid from the building plumbing system, a control valve having an inlet and an outlet in fluid communication with each other and the control valve outlet in fluid communication with the faucet housing outlet, and the split stem cartridge disposed in fluid communication with said control valve within said faucet housing in between the faucet inlet and the faucet outlet. Further described is a method of preventing pipes in a building plumbing system from bursting, comprising the step of disposing a fluid control valve system for use with a fluid having a pressure in fluid communication with the building plumbing system

that includes a faucet housing having a handle and an outlet for delivering fluid from the building plumbing system, a control valve having an inlet and an outlet in fluid communication with each other and the control valve outlet in fluid communication with the faucet housing outlet, and the split stem cartridge disposed in fluid communication with said control valve within said faucet housing in between the faucet inlet and the faucet outlet.

[0007] In yet another aspect, a pipe-burst-prevention fluid control valve cartridge for fluid having a pressure is described that includes a body having a stem end, a valve end capable of being held in fluid communication with a fluid control valve of a faucet in a plumbing system, an exterior surface, and a generally cylindrical interior surface, a generally cylindrical split stem, at least a portion of which is configured to be rotatably held within the interior of said body, having a spindle configured to receive a faucet handle that includes an upper stem sized so as to be received within the interior surface of said body and configured to be capable of being rotated when the spindle is turned, and having an upper end adjacent to the spindle and a lower end generally opposite the upper end and adjacent to the cartridge body valve end with the upper end toward the cartridge body valve end with an interior cavity extending therefrom into the upper stem toward the upper stem upper end, a lower stem having an upper end adjacent to the lower end of the upper stem with a piston affixed on the lower stem upper end and a lower end generally opposite the lower stem upper end, the lower stem configured to be slidably received into and rotatably held in the interior of said cartridge body to allow it to rotate when the spindle is rotated and prevent the flow of fluid from the inlet to the outlet when the spindle is rotated to its closed position and the pressure of the fluid is less than the predetermined pressure, and allow the flow of fluid through the outlet when the spindle is rotated to its open position or the spindle is in its closed position and the pressure of the fluid equals or exceeds the predetermined pressure, and a spring held in the cavity by the piston that is capable of compressing when the pressure of the fluid in the valve equals or exceeds the pre-determined pressure, wherein the piston, cavity and spring have diameters such that the cavity can receive slidably the spring and the piston, and lengths such that the lower end of the upper stem and the upper end of the lower stem form a gap when the spring is fully extended, and a cartridge base held in said cartridge body at its valve end having an opening adjacent to the lower end of the lower stem. In another aspect, a compact, pipe-burst-prevention control valve system for use with fluid having a pressure in a building plumbing system is described that includes a faucet housing having a handle and an outlet for delivering fluid from the building plumbing system, a control valve having an inlet and an outlet in fluid communication with each other and the control valve outlet in fluid communication with the faucet housing outlet, and the cartridge disposed in fluid communication with said control valve within said faucet housing in between the faucet inlet and the faucet outlet. Further described is a method of preventing pipes in a building plumbing system from bursting, comprising the step of disposing the fluid control valve system having the split stem cartridge for use with a fluid having a pressure in fluid communication with the building plumbing system.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a cross-sectional view taken along the longitudinal centerline of one embodiment of a pressure relief valve according to the disclosure;

[0009] FIG. 2 is a cross-sectional view taken along the longitudinal centerline of an alternative embodiment of a pressure relief valve according to the disclosure;

[0010] FIG. 3 is a cross-sectional view taken along the longitudinal centerline of a further embodiment of a pressure relief valve according to the disclosure;

[0011] FIG. 4 is a perspective view of one embodiment of a valve system having a relief valve shown in FIG. 1, 2 or 3, according to the disclosure;

[0012] FIG. 5 is a flow diagram depicting the flow of fluid through a valve system according to FIG. 4;

[0013] FIG. 6 is a perspective view of an alternative embodiment of a valve system according to the disclosure;

[0014] FIG. 7 is a flow diagram depicting the flow of fluid through a valve system according to FIG. 6;

[0015] FIG. 8 is an exploded perspective view of another embodiment of a valve system according to the disclosure;

[0016] FIG. 9 is a cross-sectional view taken along the longitudinal centerline of one embodiment of a control valve cartridge according to the disclosure;

[0017] FIGS. 10A, 10B and 10C depict cross-sectional views of the cartridge embodiment shown in FIG. 9 with the split stem in different positions;

[0018] FIG. 11 is a cross-sectional view taken along the longitudinal centerline of another embodiment of a control valve cartridge according to the disclosure;

[0019] FIGS. 12A and 12B depict cross-sectional views of the cartridge embodiment shown in FIG. 11 with the split stem in different positions;

[0020] FIG. 13 is a perspective view of the lower stem of the cartridge embodiment shown in FIGS. 11, 12A and 12B; and

[0021] FIG. 14 is an exploded perspective view of another embodiment of a valve system according to the disclosure.

DETAILED DESCRIPTION

Relief Valve

[0022] A pressure relief valve is described that is configured so that it can be installed in an existing building plumbing system to allow fluid to flow through the relief valve to protect the pipes, one or more valve control systems used in faucets, or other fixtures installed in the plumbing system from bursting when the fluid in the pipes, valve systems or other fixtures in the system freezes. The relief valve has a generally tubular and longitudinal body having an interior defining a generally longitudinal passageway through which fluid may flow. The body also has an inlet into which fluid from a high-pressure source is received, an outlet through which fluid at a lower pressure exits the relief valve, and an exterior surface that is generally longitudinal. The relief valve also has a valve seat disposed on the interior surface of the body between the inlet and the outlet, a flow limiter disposed on the interior surface of the body between the valve seat and the outlet and extends into the passageway, and a valve seat sealing portion disposed in fluid communication with the valve seat between the inlet and the flow limiter. The valve seat sealing portion is configured to seal against the valve seat so that the flow of fluid through

the valve seat and, accordingly, the passageway, is prevented when the fluid has a pressure that is equal to or less than a predetermined pressure, and not seal against the valve seat, thereby allowing fluid to flow through the valve seat and passageway, when the fluid has a pressure that is great than the predetermined pressure.

[0023] The valve seat sealing portion can be generally transverse or parallel to the exterior surface of the body. In the embodiment in which the valve sealing portion is generally transverse to the exterior surface of the relief valve body, the valve seat is generally parallel to the exterior surface of the body. In the embodiment in which the valve sealing portion is generally parallel to the exterior surface of relief valve body, the valve seat is generally transverse to body exterior.

[0024] An embodiment of the pressure relief valve according to the disclosure that has a valve seat sealing portion that is generally transverse to the exterior surface of the body is the spring-activated pressure relief valve embodiment 100 having a valve seat sealing portion 102 that has a spring 104, and a generally tubular longitudinal body 106, shown in FIG. 1. In this embodiment, body 106 has an exterior surface 108 to which valve seat sealing portion 102 is attached generally transverse to body exterior surface 108.

[0025] Valve seat sealing portion 102 may be attached using conventional means such as soldering or using screws 110. The valve seat sealing portion 102 also is preferably attached to exterior surface 108 using a seal or gasket 112 to prevent the flow of fluid between valve seat sealing portion 102 and exterior surface 108.

[0026] Body 106 has an interior surface 114 that defines a generally longitudinal passageway 116 extending through the interior surface 114 of body 106 through which fluid may flow. The body 106 also has an inlet 118 into which fluid from a high-pressure source in a building plumbing system is received, and an outlet 120 through which fluid flows through passageway 116 out of relief valve 100.

[0027] Relief valve 100 also has a valve seat 122 disposed on interior surface 114 of body 106 between inlet 118 and outlet 120 and configured to be capable of allowing fluid to flow from inlet 118 to outlet 120 through passageway 116. Valve seat 122 can be any shape but preferably, for ease of manufacturing, it has a generally cylindrically shaped opening through which fluid may flow. It preferably is constructed integrally with valve body 106. Alternatively, it may be comprised of a layer of a material that is different from the body 106 applied to interior surface 114 and glued or molded in place.

[0028] Relief valve 100 further has a flow limiter 124 disposed on interior surface 114 between valve seat 122 and outlet 120 and extending into passageway 116. The flow limiter 124 is sized such that the flow of fluid from inlet 118 to outlet 120 is reduced from a normal flow of fluid in a building plumbing system which is from about 0.5 to about 2.5 gallons per minute (“GPM”) at a normal fluid pressure which for a domestic building plumbing system for water is typically about 40 pounds per square inch (“PSI”). Preferably, flow limiter 124 is sized to reduce the flow of fluid from inlet 118 to outlet 120 by about 50% to about 80% of the normal flow, or from about 0.25 to about 1.25 GPM to about 0.10 to about 0.50 GPM. More preferably, flow limiter 124 is sized to limit the flow of fluid through relief valve 100 to less than about 0.4 GPM.

[0029] Flow limiter 124 may be comprised of a single circular opening. In this case, it can have a diameter of less than about 0.05 inches or an opening area of about 0.0020 square inches or less. Preferably, flow limiter 124 has a diameter of between about 0.015 inches to about 0.05 inches or an area of between about 1.77×10^{-4} to about 0.0020 square inches.

[0030] Preferably, flow limiter 124 may be comprised of multiple smaller, circular orifices. A flow limiter 124 comprised of multiple smaller, circular orifices reduces the chance that the flow limiter will become clogged. Where flow limiter 124 is comprised of multiple orifices, the sum of area of the orifices equals approximately that for a flow limiter having a single circular opening.

[0031] Valve seat sealing portion 102 has a movable shaft 126 on which a valve seat holder 128 which is disposed at one end closest to and capable of being held against valve seat 122. Valve seat sealing portion 102 has a longitudinal center 130 which extends through it and spring holders 132 and 134 each with centers 136 and 138, respectively, along centerline 130. Shaft 126 is disposed through spring holder center 138, through spring 104, and spring holder center 136. At least one of spring holders 132 or 134 may have threads 140, shown disposed on the inside center of holder 132, and shaft 126 may have threads 142 such that shaft 126 may be rotated to travel inward or outward, thereby adjusting the force of spring 104 to apply a greater or lesser force on shaft 126 than the force of spring 104 alone would apply to shaft 126 without adjustment, and transferring that force to valve seat holder 128 and valve seat 122 to close relief valve 100 and prevent fluid from flowing through it. FIG. 1 shows the interior center 136 of spring holder 132 and the exterior of shaft 126 as having threads on each 140 and 142, respectively.

[0032] Valve seat sealing portion 102 has a bonnet 144 which is attached sealably to exterior surface 108 of relief valve 100 over centerline 130. Bonnet 144 has a center 146 which aligns with centerline 130, and an interior surface 148 opposite valve seat holder 128.

[0033] As shown in FIG. 1, bonnet 144 holds spring holder 134 against bonnet interior surface 148 on one end opposite valve seat 122 under the force of spring 104 held at the other end adjacent to valve seat 122 by spring holder 132. Valve seat sealing portion 102 has a collar 150 with a center 152 upon which spring holder 132 rests against the force of spring 104. Where valve seat sealing portion 102 has two spring holders, as shown in FIG. 1, bonnet 144 holds spring holder 134 against bonnet interior surface 148 under the force of spring 104 with spring holder 132 holding spring 104 at its other end against collar 150. Valve seat sealing portion 102 need not have spring holder 134, but rather spring 104 may rest against bonnet interior 148. Alternatively, valve seat sealing portion 102 need not have spring holder 132, in which case spring 104 may rest on collar 150 at one end and spring holder 134 at the other end, and spring holder 134 may have threads on its interior and movable shaft 126 may have threads disposed on its exterior adjacent to the threads on the interior of spring holder 134 thereby allowing the force applied to valve seat holder 128 against valve seat 122 to be adjusted.

[0034] Valve seat holder 128 has a flange portion 154 which is disposed adjacent to valve seat 122 and configured to be capable of holding and sealing valve seat 122. Valve seat holder 128 further has a rod portion 156 extending away

from flange portion 154 along centerline 130 through collar center 152 and abutting shaft 126.

[0035] In the embodiment shown in FIG. 1, shaft 126 extends through spring holder 132, through its center 136, through spring 104 and spring holder 134 through its center 138. Shaft 126 may exit bonnet 144 through its center 146 opposite valve seat holder 128 to allow spring holder 132 or 134 to be adjusted by rotating it. Shaft 126 also may have knob 158 to allow an end user to rotate shaft 126 more easily. Further, knob 158 may have knurls on its exterior, which further facilitate gripping and rotation of knob 158.

[0036] Bonnet 144 may be secured sealably to exterior surface of relief valve body 108 using convention means. In FIG. 1, bonnet 144 has a flange 160, and bonnet 144 is secured to exterior surface 108 with seal 112 in between bonnet flange 160 and exterior surface 108 using screws 110. In addition, movable shaft 126 preferably has a seal between it and bonnet 144 at its center 146 where it exits so that fluid cannot flow between shaft 126 and bonnet center 146. Similarly, collar 150 should have a seal between it and bonnet 144 and rod portion 156 so that fluid cannot flow through flow limiter 124 unless pressure relief valve 100 has been activated.

[0037] Relief valve 100 may be made using convention plumbing manufacturing methods. Relief valve 100 is used by installing it in a building plumbing system either by soldering it into a plumbing line, or by manufacturing it with threaded inlet 118 and outlet 120 and using threaded couplings to secure it in fluid communication with the building plumbing system.

[0038] Relief valve 100 inlet 118 receives fluid from a hot-or cold-water source in a domestic plumbing system. From inlet 118, the fluid can flow through passageway 116 of body 108 and valve seat 122 if the pressure of the fluid is greater than a predetermined pressure. The predetermined pressure is the force required to activate or lift valve seat holder 128 off valve seat 122 to deliver fluid through valve seat 122 to outlet 120 and the delivery end of a faucet, other valve, or other fixture in the building plumbing system.

[0039] Flow limiter 124 restricts the flow of fluid through relief valve 100 so that the pressure of the fluid is relieved when the fluid in the line freezes but will flow at less than the rate of flow through an open valve in the building plumbing system. Thus, flow limiter 124 protects the surroundings in case the flow of fluid has the potential to cause damage.

[0040] Spring 104 has a compressive force such that when divided by the area of the open area of valve seat 122 which is in contact with seat holder 128 is approximately equal to or less than the predetermined pressure. The activation pressure is the compressive pressure required to activate relief valve 100 and overcome the force required to compress spring 104 so that fluid may flow through relief valve 100. The predetermined pressure can be preset by selecting spring 104 that has a compressive force that will allow it to be compressed at the pressure desired to activate relief valve 100. The predetermined pressure also can be preset by selecting spring 104 to have a desired compressive force such that when one or both spring holders 132 and 134 are present the activation pressure of relief valve 100 can be preset to be higher than the compressive force of the spring 104 alone. In either case, spring 104 has a force such that it is capable of applying the desired pressure on seat holder 128, which in turn applies pressure to the valve seat 122 and

closes relief valve 100 when the pressure of the fluid in the building plumbing system is less than or equal to the pressure that will burst one or more pipes, valves or fixtures in the building plumbing system, and activates or opens relief valve 100 when the pressure of the fluid in the building plumbing system is greater than or equal to the pressure that will burst one or more pipes, valves or fixtures in the building plumbing system.

[0041] A typical water control valve used in domestic plumbing is estimated to have a break strength of around 4,000 PSI. A pressure that would break a typical copper plumbing pipe is approximately 1,000 PSI, depending on the wall thickness and method used to manufacture the pipe. Plastic (PEX) pipe will typically burst at around 500 PSI. Normal water pressure in a domestic plumbing system is typically less than 100 PSI, and, more typically, around 40 PSI.

[0042] Preferably, spring 104 has a compressive pressure that is less than the pressure at which the weakest component in the plumbing system will burst. The compressive pressure required to compress spring 104 is the force required to compress the spring divided by the area of the valve seat opening which is in contact with the seat holder when the relief valve is closed. If relief valve 100 has at least one spring holder 132 or 134 that is adjustable on shaft 126, then spring 104 can have a lower compressive pressure than required to activate relief valve 100 if holder 132 or 134 may be adjusted to increase the pressure required to compress the spring as adjusted desired force.

[0043] Spring 104 in relief valve 100 for use in a plumbing system with plastic (PEX) piping preferably has a compressive force such that relief valve 100 will have an activation pressure of less than about 500 PSI. More preferably, spring 104 used in relief valve 100 for a plumbing system with plastic (PEX) piping has a compressive force such that relief valve 100 has an activation pressure of between about 200 and about 395 PSI, and, even more preferably, between about 300 and about 395 PSI.

[0044] For a building plumbing system that has no plastic PEX piping but rather only copper piping, spring 104 preferably has a compressive force such that relief valve 100 has an activation pressure of less than about 1,000 PSI. More preferably, spring 104 used in relief valve 100 in a plumbing system with only copper piping has a compressive force such that relief valve 100 has an activation pressure of between about 400 and about 995 PSI, and, even more preferably, between about 500 and about 995 PSI.

[0045] In another spring-activated embodiment according to the disclosure, the valve seat sealing portion can be disposed within the passageway generally parallel to the exterior surface of the body, and the valve seat disposed in the passageway generally transverse to the interior and exterior surfaces of the relief valve body, as shown in FIG. 2. Relief valve 200 has a valve seat sealing portion 202 with a spring 204, and a generally tubular elongate body 206 with an exterior surface 208 and an interior surface 210 defining a passageway 212 that extends through body 206 through which fluid may flow. Valve seat sealing portion 202 is disposed generally parallel to exterior and interior surfaces 208 and 210 within passageway 212. Relief valve 200 further has an inlet 214 capable of receiving fluid from a source or other plumbing fixture at relatively high pressure, such as that from a domestic plumbing system cold- or hot-water source, and an outlet 216 capable of receiving

fluid at a lower pressure from the inlet **214** after the fluid has traveled through passageway **212** and exiting relief valve **200** for delivery to a faucet, other valve or other fixture in a building plumbing system, and a valve seat **218** disposed in between inlet **214** and outlet **216** generally transverse to passageway **212**.

[0046] Valve seat sealing portion **202** has a bonnet **220** attached over valve seat **218** so that fluid flowing through valve seat **218** passes through bonnet **220**. Bonnet **220** may be soldered in place. Or bonnet **220** may have a flange **222** and be attached with a retainer **224** and screws to flange **222**. Preferably, bonnet **220** is sealably attached so that fluid cannot flow between bonnet flange **222** and interior surface **210** using a seal **226** therebetween.

[0047] Bonnet **220** holds spring **204** against a ball **228**, which in turn is forced down against valve seat **218** by spring **204**, thereby sealing relief valve **200**. Ball **228** is generally round for ease of manufacturing but could have other shapes. Valve seat **218** is generally round for receiving a round ball for ease of manufacturing but could have other shapes so long as ball seats in valve seat to seal it. Valve seat **218** and ball **228** are sized to seal with each other and prevent the flow of fluid through valve **200** when ball **228** is forced against valve seat **218** by spring **204**.

[0048] Valve seat **218** may be comprised of the same material of which body **206** is comprised. Valve seat **218** could be made of a material that is different from the material used to make body **206**, like valve seat **122** of relief valve **100**.

[0049] Relief valve **200** also has a flow limiter **230** which allows fluid to flow through relief valve **200** once activated but restricts the flow to less than the normal flow of fluid through a building plumbing pipe. Flow limiter **230** is disposed in bonnet **220** distal to bonnet flange **222**. Flow limiter **230** may have the features and be sized like flow limiter **124** of relief valve **100**.

[0050] The force required to allow fluid to flow through relief valve **200**, in other words its activation pressure, can be preset by selecting spring **204** with a compressive force that will exert the desired pressure on ball **228** to prevent fluid from flowing through valve seat **218**, keeping relief valve **200** closed until the force of the fluid exceeds the compressive force of spring **204**. Spring **204** has a compressive force as described relative to spring **104** and relief valve **100** that is less than the force which would burst the weakest component in the building plumbing system.

[0051] The activation pressure of relief valve **200** could be increased above the compressive force of spring **204** by including spacers within bonnet **220** or using a bonnet **220** sized to hold spring **204** in a compressed state. Doing so would increase the force that spring **204** would apply to ball **228**, which would in turn increase the force that would be required to open relief valve **200**.

[0052] Another relief valve embodiment according to the disclosure is shown in FIG. 3. Relief valve **300** has a body **302** with an exterior surface **304** and an interior surface **306** defining a generally longitudinal passageway **308** through which fluid can flow from an inlet **310** for receiving fluid at a relatively high pressure from a fluid source such as hot-or cold-water or other plumbing fixture to an outlet **312** where fluid exits relief valve **300** at a lower pressure than it enters inlet **310** for delivery to a faucet or other plumbing pipe or fixture, a valve seat **314** disposed on interior surface **306** and a rupture or burst disk **316** disposed over valve seat **314**

across passageway **308** such that disk **316** is capable of preventing the flow of fluid through the passageway **308** when the fluid is below a predetermined pressure.

[0053] Rupture disk **316** is comprised of a material having a break strength which bursts at the predetermined pressure. Materials that can be used for rupture disk **316** are metal, plastic, such as polyethylene or polypropylene, or elastomers, such as butyl rubber. Stainless steel or other non-corroding material is preferred. Rupture disk **316** can be flat or concave. It also can have one or more scores or areas that are thinner or configured such that one or more of them break at a pressure that is less than the break strength of the disk material surrounding the one or more scores.

[0054] The material used for the rupture disk **316** without a score preferably has a break strength that is less than the pressure at which the weakest component in the building plumbing system in which relief valve **300** is disposed, as described relative to relief valve **100**. If rupture disk **316** has a score in it, a material having a higher break strength could be used for rupture disk **316** so long as the score breaks at a break strength that is less than the break strength of copper pipe if a plumbing system with only copper pipe is used, or less than the break strength of plastic pipe if the plumbing system has some plastic pipe. In this way, relief valve **300** would remain closed until the force of the fluid in the inlet **310** exceeds rupture disk **316** break strength at which point it would break, relieving the pressure in the line so that other pipes, valves, or fixtures in the plumbing system do not break.

[0055] Rupture disk **316** can be held in place by soldering, screwing, gluing, or using other conventional means to hold it into place. Rupture disk **316** can be held in body **302** by retainer **318**. Alternatively, relief valve **300** can have an inlet end **320** and a separate outlet end **322** each having threads that allow them to be threaded into one another with rupture disk **316** held in between.

[0056] More specifically, as shown in FIG. 3, relief valve **300** can comprise separate inlet and outlet ends **320** and **322** each with threads **324** and **326**, inlet end threads **324** disposed on the interior of inlet end **320** adjacent to threads **326** on outlet end **322** so that outlet end **322** can be screwed into inlet end **320**. Relief valve **300** has rupture disk **316** disposed in inlet end **320** across passageway **308** with retainer **318** disposed between rupture disk **316** and outlet end **322**, securing rupture disk **316** in place against valve seat **314** by screwing outlet end **322** into inlet end **320**.

[0057] Additionally, rupture disk **316** may have a flange **328** on its outer edge sized to be capable of being held by retainer **318** against valve seat **314**. Flange **328** preferably is molded into rupture disk **316**.

[0058] Alternatively, inlet end **320** and outlet end **322** can be threaded and configured so that inlet end **320** can be received threadably into outlet end **322** with retainer **318** held between outlet end **322** and rupture disk **316**, and rupture disk **316** held against a valve seat inside outlet end **322**.

[0059] Relief valve **300** may also have a flow limiter **330** disposed on interior surface **306** through which fluid can flow in a restricted manner from high-pressure inlet **310** to low-pressure outlet **312** when the pressure of the fluid in inlet **310** is high enough to burst rupture disk **316**. Flow limiter **330** is sized and shaped like flow limiters **124** and **230** of relief valves **100** and **200**, respectively. In FIG. 3,

flow limiter **330** is disposed on the interior of outlet end **322** between outlet **312** and rupture disk **316**.

[0060] Use of a flow limiters **124**, **230** and **330** is preferred because, when relief valves **100**, **200** and **300** open under pressurized conditions, they will prevent a large flow of fluid from flowing through the pipe which could overflow a tub or sink or other fixture and damage the building in which the relief valve is used.

[0061] One or more relief valves **100**, **200** or **300** could be connected at two points on a pipe without any control valve or other fixture or plumbing component in between but near a control valve or other fixture or component. By doing so, the control valve or other plumbing fixture is protected from freezing. Alternatively, a relief valve **100**, **200** or **300** could be connected to two pipes, one pipe supplying a control valve for a faucet, or other fixture, such as a shower mixing and control valve system, and one pipe connected to the downstream end of the control valve faucet or fixture. In doing so, relief valve **100**, **200** or **300** relieves pressure and protects the pipe, control or other valve, or plumbing system component.

[0062] The predetermined pressure for activation of pressure relief valves **100**, **200** and **300** is preferably less than about 1000 PSI for a building plumbing system with entirely copper piping. More preferably, the predetermined activation pressure used in relief valves **100**, **200** and **300** installed in a building plumbing system with all copper piping will be between about 400 and about 995 PSI, and, even more preferably, between about 500 and about 995 PSI.

[0063] If the building plumbing system includes plastic PEX piping, then the predetermined activation pressure for relief valves **100**, **200** and **300**, preferably, should be less than about 500 PSI. If it is unknown what piping is used in a building plumbing system, then pressure relief valve **100**, **200** or **300** used in the system preferably should have a predetermined activation pressure of less than about 500 PSI, and more preferably between about 200 and about 395 PSI, and, even more preferably, between about 300 and about 395 PSI.

[0064] Spring **104**, **204** and rupture disk **316** preferably have activation pressures of less than the burst pressure of the weakest component in the building plumbing system. When spring **104**, **204** and rupture disk **316** have activation pressures that are less than the weakest component in the building plumbing system, relief valves **100**, **200** and **300** will activate and relieve the pressure in the building plumbing system before a burst results.

Control Valve System with a Relief Valve

[0065] A pipe burst prevention valve system for use in a building plumbing system is described having a control valve and a relief valve **100**, **200** or **300**. For example, as shown in FIG. 4, pipe burst prevention valve system **400** according to the disclosure has a mixing and diverter valve **402** on which a faucet housing with a handle would be placed. FIG. 4 shows valve system **400** having two inlets from two high-pressure sources **404A** and **404B**, typically, cold-and hot-water lines, respectively, in fluid communication with control valves **406A** and **406B**, one for controlling cold water from cold-water line **404A** and one for controlling hot water from hot-water line **404B**, respectively, which each also would have a handle on the faucet housing for turning each control valve **404A** and **404B** on or off, two outlets **408A** and **408B**, and two bypasses **410A** and **410B**,

each having a pressure relief valve **412A** and **412B** according to the disclosure, such as one of pressure relief valve **100**, **200** or **300**, an inlet **414A** and **414B**, with inlet **414A** in fluid communication with control valve inlet **404A** and inlet **414B** in fluid communication with control valve inlet **404B**, and an outlet **416A** and **416B**, with outlets **416A** and **416B** in fluid communication with outlet **408B**.

[0066] Typically, for a valve system like that shown in FIG. 4, one outlet **408A** would deliver water to a bathtub faucet, and the other outlet **408B** would deliver water to a shower head, either at the same time or separately, depending upon mixing and diverting valve **402** and the position of the handle controlling it. Bypasses **410A** and **410B** will allow water to flow through them when the pressure in inlets **404A** and **404B** exceeds the force required to activate relief valve **412A** or **412B** thereby bypassing control valves **406A** and **406B** and mixing valve **402** and protecting them and the pipes in the building plumbing system from freezing. Bypass **410A** or **410B** could be in fluid communication with outlet **408A** instead of **408B**, or one could be connected to outlet **408A**, and the other bypass could be connected to outlet **408B**.

[0067] Spindles **418A** and **418B** are for attaching cold-and hot-water handles to control valves **406A** and **406B**, respectively, so that the valves can be operated. Spindle **420** is for attaching a mixing and diverting valve handle to the valve **402** to operate it.

[0068] Often, only the cold-water control valve inlet pipe is at risk of freezing in a domestic plumbing system, and not the hot-water control valve pipe inlet. In this case, a valve system could have only one bypass disposed to bypass the cold-water control valve and protect it from freezing. However, where there is a risk that the building plumbing system's hot water heating system could fail, or where the building heat could fail or both and there is a risk that the pipes will freeze, then a valve system with more than one bypass is preferred.

[0069] Preferably, valve system **400** has spring-operated relief valve embodiments **100** and **200** as relief valves **412A** and **412B** in bypasses **410A** and **410B** because they will continue to protect a building plumbing system after a freezing event. Once a frozen pipe thaws and the pressure of the building plumbing system returns to normal, relief valves **100** and **200** close and water no longer would flow through them, but they will open again if the pipes freeze again. Relief valve **300**, however, is preferred where space and cost are limiting factors.

[0070] Bypasses **410A** and **410B** according to the disclosure can also have relief valves **100**, **200** or **300** within a Y-type access pipe having an access port which would allow the relief valves **100**, **200** or **300** to be inspected and serviced with minimal downtime for the plumbing system. Relief valve **100**, **200** or **300** needing to be replaced could be removed or replaced using the access port. This is especially important for relief valve **300** which will no longer function to relieve pressure in the pipe once its rupture disk **316** has burst. Therefore, relief valve **300**, once ruptured, will need to be replaced to permit valve system **400** to operate as described and to continue to protect the plumbing system or its pipes or other components from freezing.

[0071] FIG. 5 is a flow chart showing the flow of fluid through a valve system **400**, shown in FIG. 4. In FIG. 5, fluid flows through the pipe burst prevention valve system **400** from two inlets from two high-pressure fluid sources **404A**

and 404B, typically, cold-and hot-water lines, respectively, in fluid communication with control valves 406A and 406B, one for controlling cold water from cold-water line 404A and one for controlling hot water from hot-water line 404B, respectively, to outlets 408A and 408B, depending upon the position of the mixing and diverter valve 402. Fluid flows through bypasses 410A and 410B, each having a pressure relief valve 412A and 412B according to the disclosure such as one of pressure relief valve 100, 200 or 300, when the pressure of the fluid in inlets 404A or 404B is at or above the activation force of relief valve 412A or 412B thereby bypassing control valves 406A and 406B and mixing valve 402 and protecting them and the pipes connecting them to the building plumbing system from freezing by allowing fluid to flow through outlet 408A. In FIG. 5, outlet 408A could be a bathtub faucet and outlet 408B could be a shower faucet.

[0072] FIG. 6 is a perspective view of an alternative embodiment of a valve system according to the disclosure. Valve system 500 shown in FIG. 6 comprises two control valves for controlling the flow of fluid 502A and 502B from an inlet from a high-pressure cold-water source 504A and from a high-pressure hot-water source 504B, respectively, and each such control valve 502A and 502B having outlets 506A and 506B, respectively, which combine into one control valve system outlet 508, such as a faucet for a bathtub or shower. Valve system 500 further has two bypasses 510A and 510B each having a pressure relief valve 512A and 512B according to the disclosure, respectively, such as one of embodiments 100, 200, or 300 shown in FIG. 1, 2 or 3, disposed in fluid communication with control valve inlets 504A and 504B and control valve outlets 506A and 506B. System 500 also has spindles 514A and 514B configured to receive handles to aid in turning control valves 502A and 502B on and off.

[0073] FIG. 7 is a flow diagram showing the flow of fluid through valve system 500, which has control valves 502A and 502B for controlling the flow of fluid from high-pressure inlets 504A and 504B, respectively, and flowing out to a combined control valve system outlet 508, and two bypasses 510A and 510B disposed in fluid communication with inlets 504A and 504B, respectively, and control valve system outlet 508, each bypass 510A and 510B having a relief valve 512A or 512B, according to the disclosure. System 500 could have only one bypass, preferably on the cold-water source side of system 500, instead of two as shown.

[0074] Another embodiment according to the disclosure is a control valve system that includes another low-pressure delivery pipe end, not shown in FIG. 6 or 7. In yet another embodiment according to the disclosure a diverter that may be operated between one or more deliver end pipes, such as where a shower has multiple shower heads, which diverts the water from a first low-pressure delivery pipe to a second low-pressure delivery pipe or more low-pressure delivery pipes, and back again. For example, such a diverting valve could be advantageously used to divert water being delivered from one location to another or to two or more locations, such as a bathtub to a shower or multiple shower heads.

[0075] In yet another embodiment, the system can include a mixing valve that mixes and moderates hot-and cold-water delivered to the mixing valve. Such a mixing valve also can be pressure controlled. Such a mixing valve can also have a

diverting function. Such a mixing valve also may not have a diverting function and be used to deliver water to a diverter valve.

[0076] FIG. 8 is an exploded perspective view of another embodiment of a control valve system according to the disclosure. In FIG. 8, valve system 600 has a cartridge 602 that receives fluid from fluid sources through inlets 604A and 604B and mixes them as well as controls their volume and allows the fluid to exit from a single outlet 606. Valve system 600 also has two bypasses, 608A and 608B, in fluid communication with inlets 604A and 604B, respectively, and control valve system outlet 606. Valve system 600 also is shown having an aerator 610. Valve system 600 has faucet housing 612 which is disposed over cartridge 602 and out of which fluid flows through outlet 606 and has a handle 614 for controlling fluid flow from inlets 604A and 604B, center lines of which are shown in dashed lines, through cartridge 602. Bypasses 608A and 608B have a relief valve according to the disclosure, such as relief valve 100, 200 or 300.

[0077] In addition, a valve system according to the disclosure could include additional outlets, additional mixing, control valves, such as thermostatic and pressure control valves, as well as a bypass with a pressure relief valve according to the disclosure, such as relief valve 100, 200 or 300. Further, valve system 400, 500, and 600 could also include in bypasses 410A, 410B, 510A, 510B, 608A and 608B a Y-type access port disposed to allow relief valves 412A, 412B, 512A, 512B, 608A, and 608B to be serviced.

[0078] Valve systems 400, 500, and 600 can be made by soldering, screwing, or clamping the components described in relation to them to pipes in fluid communication with them. Alternatively, they can be made by molding the valves and bypasses into an assembly and attaching the assembly using fittings or soldering the assembly to pipes in fluid communication with the building plumbing system. Relief valves and cartridges may be made removable and replaceable by threading or using O-rings. Preferably, a valve system according to the disclosure is molded such that bypasses are integral and generally parallel to the valve that they bypass so that they are compact enough to be installed in a two-by-four wall, typical in residential dwelling walls, like conventional shower or tub wall mixing, diverter and control valves can be installed, yet the valve system shall be molded such that the relief valve and any cartridge shall be removable so that they can be replaced when necessary. The valve system according to the disclosure advantageously can also be installed in a wall on the exterior of a residential dwelling because it is protected against freezing.

Control Valve Cartridge with a Split Stem and a Washer

[0079] Pipe bursts due to freezing also can be prevented by using a compact, pipe-burst-prevention fluid control valve cartridge according to the disclosure for use with a fluid control valve faucet for controlling the flow of fluid having a pressure in a plumbing system. One embodiment of such a cartridge is a generally cylindrical control valve cartridge 700, as shown in FIG. 9, having a washer 702 and a split stem 704. More particularly, cartridge 700 has a stem end 706 opposite a valve end 708, a cartridge body 710 that has exterior and interior surfaces 712 and 714, respectively, with upper and lower interior threads 716 and 718, upper interior threads 716 toward cartridge upper stem end 706 and lower interior threads 718 toward cartridge valve end

708, and cartridge body **710** further has exterior threads **720** toward cartridge valve end **708** configured to be capable of being screwed into faucet housing to hold cartridge **700** in place in the faucet housing. Cartridge **700** has split stem **704** having an upper stem **722** with a generally cylindrical exterior surface **724** with threads **726** thereon adjacent to cartridge interior surface lower end threads **718**, and a shoulder **766** formed between stem threads **726** extending beyond upper stem exterior surface **724**.

[0080] Cartridge **700** further can have a packing **730** disposed around upper stem exterior surface **724** on upper stem thread shoulder **766**, and a packing nut **732** having an interior surface **768** configured to be capable of rotatably receiving upper stem **722**, threads on its exterior surface **734** configured to be capable of being threaded into upper interior cartridge body threads **716**, and a lower end **770** capable of resting on stem thread shoulder **766** when the spindle **740** is turned to its fully open position. Cartridge **700** further can have a stop **772** disposed on shoulder **766** of upper stem threads **726** and, in this case as shown in FIG. 9, packing **730** is disposed on stop **772** and packing nut **732** is configured to screw into body **710** with packing nut lower end **770** abutting packing **730**, and packing **730** abutting stop **772**.

[0081] Upper stem **722** has an upper end **736** toward the cartridge stem end **706** and a lower end **738** toward the valve end **708** of the cartridge **700**. The upper stem **722** has a spindle **740** on which a handle for a faucet may be attached. Upper stem **722** also has a generally cylindrical interior cavity **742** extending from the upper stem lower end **738** toward the upper stem upper end **736**, a calibrated spring **744** disposed in cavity **742** toward the upper stem upper end **736**, and a clip **762** attached to upper stem lower end **738** adjacent to cavity **742**. Preferably, upper stem lower end **738** has at least two clips **762** opposing one another on opposite sides of cavity **742**.

[0082] Cartridge split stem **704** also has a lower stem **728** toward cartridge valve end **708** with generally cylindrical upper end and lower ends **746** and **748**, respectively, the lower end **748** having a diameter that is larger than the diameter of upper end **746** thereby forming a shoulder **750**. Lower stem upper end **746** has a diameter that is sized so that it is capable of sliding into and being received by cartridge body interior **714** at its valve end **708**. Lower stem **728** also has a piston **752** having an upper end **754**, a lower end **756** attached to the upper end of the lower stem **746** and having a flange **758** adjacent to piston lower end **756**, and a retaining screw **760** attaching washer **702** to the lower stem lower end **748**. Piston **752** can be formed integrally with the lower stem lower end **748** or it can be screwed or soldered onto it. Alternatively, two or more of lower stem upper end **746**, lower stem lower end **748** and piston **752** could be molded together as one piece.

[0083] Piston **752** has a size such that it is capable of slidably being received within cavity **742** and holding a calibrated spring **744** in cavity **742** at its upper end **754**. Piston **752** is capable of sliding between a fully extended position toward valve end **708**, held by a piston retaining clip **762** on flange **758** and thereby defining a gap **764** between lower stem upper end **746** and upper stem lower end **738**, and fully compressed position, wherein lower stem **728** is compressed upward toward stem end **706** until shoulder **750** rests on cartridge body **710** at its valve end **708**.

[0084] Cartridge **700** can be used in single-spindle compression washer control valves, like valves **406A** and **406B**, shown in FIG. 4, and valves **502A** and **502B**, shown in FIG. 6. Cartridge **700** is used by inserting its valve end **708** into a control valve of a faucet and screwing it into the faucet so that washer **702** and valve end **708** rest on the valve seat of the faucet. Cartridge **700** may not have exterior threads **720** which screw into the valve but rather may have an O-ring that friction fits into the faucet valve. When faucet is opened and closed by turning the handle of the faucet attached to spindle **740**, upper stem **722** rotates freely within packing nut **732** and moves in threads **726** toward cartridge stem end **706** or valve end **708** on interior threads on lower valve end **718**, which raises or lowers valve end **708** of the split stem **704** off or onto valve seat to allow or prevent the flow of fluid through the valve.

[0085] Piston **752** may be configured to rotate with upper stem **722** when spindle **740** is rotated to open or close the valve yet slide longitudinally relative to upper stem **722**. In that case, piston **752** would have an elongate protrusion or key on the exterior of piston **752** extending radially outward toward and longitudinally down a portion of the length of piston **752**, and cavity **742** would have a complementary recess in it that is configured to longitudinally, slidably receive the piston key and hold it fixed relative to upper stem **722** when it is rotated thereby rotating lower stem **728** and washer **702** when spindle **740** is rotated.

[0086] When faucet handle on spindle **740** is turned to open the valve, fluid flows from the building plumbing system past washer **702** and valve seat. In FIG. 4, when spindle **418A** or **418B** are turned, washer (not shown) raises, and fluid flows past valves **406A** or **406B** and out faucet outlet **408A** or **408B** depending upon the position of mixing valve **420**. In FIG. 6, when spindle **514A** or **514B** is turned, washer (not shown) would raise, and fluid would flow past valve **502A** or **502B** and out faucet outlet **508**. In each case, where cartridges disclosed are included in a valve system of a building plumbing system, opening the valve by turning the spindle would allow water to flow from the building plumbing system water source through outlet to be used by an end user. In each case, closing the valve by turning the spindle prevents water from flowing from the building plumbing system water source through outlet to an end user under normal fluid pressure conditions but, under freezing conditions, fluid pressure increases above the predetermined pressure and water will flow to an outlet.

[0087] FIG. 10A shows cartridge **700** under normal fluid pressure conditions in a valve in its fully open position. Under these conditions, as shown in FIG. 10A, when spindle **740** has been turned to open the valve and raise upper stem **722** and washer **702** off faucet valve seat (represented by dashed line), shoulder **750** rests on cartridge body **710**. In this position, fluid can flow through the valve past washer **702**.

[0088] FIG. 10B shows cartridge **700** under normal fluid pressure conditions in a valve with cartridge **700** in its fully closed position. Under these conditions, as shown in FIG. 10B, when spindle **740** has been turned to close the valve and lower upper stem **722**, washer **702** rests on faucet valve seat (again represented by dashed line). In this position, fluid is prevented from flowing through the valve and past washer **702**.

[0089] FIG. 10C shows cartridge **700** with its stem **722** in its fully closed position but under pressurized fluid condi-

tions. Under these conditions, as shown in FIG. 10C, spindle 740 has been turned to its closed position which advances upper stem 722 in cartridge body too its extended position off of but lower stem 728 and washer 702 are pushed up off of the faucet valve seat (shown by dotted line) by pressurized fluid which exceeds the compressive force of spring 744, and activation pressure or predetermined pressure of cartridge 700, in turn pushing up piston 752 into cavity 742 allowing fluid to flow in the pipe past cartridge valve end 708 and valve seat, and through the valve.

[0090] Cartridge 700 is configured, and the lengths of cavity 742, spring 744 and piston 752 are sized so that when spindle 740 is in its fully closed position but the valve is under pressurized conditions, the fluid pressure exceeds the predetermined pressure, and lower stem lower end 748 raises off valve seat so as to allow at least some fluid to flow through the valve. Preferably, cartridge 700 is configured and the lengths of cavity 742, spring 744 and piston 752 are sized so that when spindle 740 is in its fully closed position but fluid pressure in the valve exceeds the predetermined pressure, lower stem lower end 748 raises off valve seat by an amount that is less than it raises off the valve seat when spindle 740 is turned to fully open the valve. More preferably, cartridge 700 is configured and the lengths of cavity 742, spring 744 and piston 752 are such that when spindle 740 is in its fully closed position but fluid pressure in the valve exceeds the predetermined pressure, lower stem lower end 748 raises off valve seat by a distance that allows between about 50% to about 80% less than the normal flow of fluid through the valve when the valve is open, or by a distance that allows between about 0.25 and about 1.25 GPM and about 0.10 and about 0.50 GPM to flow through the valve. Even more preferably, cartridge 700 is configured and the lengths of cavity 742, spring 744 and piston 752 are such that when spindle 740 is in its fully closed position but fluid pressure in the valve exceeds the predetermined pressure, lower stem lower end 748 raises off valve seat by a distance that allows an amount of fluid to flow through the valve that is less than about 0.4 GPM.

[0091] Spring 744 has a compressive force so that cartridge 700 is capable of relieving pressure in the building plumbing system at the desired predetermined pressure. Thus, spring 744 is capable of compressing when the pressure of the fluid in cartridge 700 exceeds predetermined pressure. Spring 744 should have a predetermined activation pressure as described above according to the type of pipes and components in the building plumbing system.

Control Valve Cartridge with a Split Stem and Ceramic Disks

[0092] Another cartridge embodiment according to the disclosure is shown in FIG. 11. FIG. 11 depicts a generally cylindrical control valve cartridge 800 for a single spindle control valve having a valve end 802 and a stem end 804 generally opposite each other along a longitudinal axis (shown in dashed line) through the center of cartridge 800, a cartridge base 806 at valve end 802, and a split stem 808. Cartridge 800 also has a body 810 that has an exterior surface 812 with an O-ring 814 on it, cartridge body exterior surface 812 being shaped to receive O-ring 814 and friction fit into a faucet valve, and an interior surface 816 that is sized to hold split stem 808 with the assistance of two O-rings 818 and 820 for holding and sealing split stem 808 to the cartridge body interior surface 816.

[0093] Split stem 808 has an upper stem 822 with a generally cylindrical exterior surface 824 with recesses 826 and 828 each sized to receive O-ring 818 and 820 and hold and seal upper stem 822 in cartridge body 810. Split stem 808 also has a lower stem 830 toward valve end 802 adjacent to upper stem 822.

[0094] Upper stem 822 has an upper end 832 adjacent to the cartridge stem end 804, a lower end 834 distal to upper stem upper end 832, and a spindle 836 on which a handle for a faucet may be attached and used to rotate spindle 836 and split stem 808 to open the valve and turn the faucet on or off. Upper stem 822 also has interior cavity 838 extending longitudinally through upper stem 822 from its lower end 834 toward upper stem upper end 832.

[0095] Lower stem 830 has an upper end 840 adjacent to upper stem lower end 834 and a lower end 842 having a ceramic control disk 844. Lower stem 830 also has piston 846 disposed on lower stem upper end 840 and extending longitudinally into cavity 838 toward stem end 804.

[0096] Piston 846 can be formed integrally with lower stem upper end 840. Alternatively, piston 846 can be screwed or soldered on to lower stem upper end 840 to affix it to lower stem 830.

[0097] Cartridge 800 also has pressure relief spring 848 disposed in cavity 838 and held in place by piston 846. Spring 848 has a compressive force that allows cartridge 800 to relieve pressure in the building plumbing system at the desired predetermined pressure. Thus, spring 848 is capable of compressing when the pressure of the fluid in cartridge 800 exceeds the predetermined pressure. Spring 848 has a predetermined activation pressure as described above according to the type of pipes and components in the building plumbing system.

[0098] Cartridge base 806 is fixedly disposed on interior surface of cartridge body 816 at its valve end 802 across a high-pressure inlet 850 and has at least one opening through which fluid may flow from a high-pressure source, such as a hot-or cold-water source. Cartridge base 806 holds a ceramic cartridge base 854 in cartridge body 810 at its valve end 802. Ceramic cartridge base 854 also has at least one opening through which fluid from opening in base 806 may flow.

[0099] Ceramic control disk 844 is disposed on lower stem 830 at its lower end 842 and forced toward ceramic cartridge base 854 by piston 846 under the force of spring 848. Ceramic control disk 844 has at least one opening configured to allow fluid to flow from inlet 850 through opening in ceramic cartridge base 854 if ceramic control disk 844 is rotated to align its opening with low-pressure outlet 860 and to prevent the flow of fluid from inlet 850 to outlet 850 if ceramic control disk 844 is rotated the opposite way thereby misaligning opening in control disk 844 with outlet 860.

[0100] Cartridge 800 further has retaining ring 862 that rotatably holds upper stem 822 in cartridge body 810 at its stem end 804 thereby holding spring 848 and piston 846 in cavity 838, and lower stem 830 against ceramic cartridge base 854. Ceramic cartridge base 854 in turn is fixedly held against cartridge base 806 in cartridge body 810 by conventional means. Retaining ring 862 and cartridge base 806 may be held in place by screwing them into place, by friction fitting them into place using O-rings, or by any other conventional means.

[0101] Piston 846, cavity 838 and pressure relief spring 848 have diameters such that piston 846 and spring 848 can

slide into cavity **838**. Piston **846**, cavity **838** and spring **848** have lengths such that a portion of piston **846** is capable of sliding into cavity **838** and holding spring **848** therein between a fully extended position, wherein lower stem **830** is held by ceramic cartridge base **854** against the force of spring **848** under normal fluid pressure conditions and thereby defining a stem gap **864** in split stem **808** between upper stem **822** and lower stem **830**, and a fully retracted position, when the pressure of the fluid at the valve end exceeds the predetermined pressure thereby compressing spring **848**, closing stem gap **864**, and defining a ceramic disk gap **866** between ceramic control disk **844** and ceramic cartridge base **854** when the pressurized fluid forces control disk **844** upward forcing lower stem **830** toward upper stem lower surface **834**, closing stem gap **864**, defining ceramic disk cap **866**, and allowing an amount of fluid to flow out of cartridge **800** through low pressure outlet **860**.

[0102] Preferably, cartridge **800** is configured and the lengths of piston **846**, cavity **838**, pressure relief spring **848**, upper end of lower stem **840** and ceramic control disk **844** are sized so that when the fluid pressure exceeds the predetermined pressure, lower stem lower end **842** raises off of ceramic base **854** so as to allow an amount of fluid to flow from cartridge inlet **850** to cartridge outlet **860** that is less than that the amount of fluid that would flow through the outlet **860** when spindle **836** is in its fully open position and the valve is fully open. More preferably, cartridge **800** is configured and the lengths of piston **846**, cavity **838**, pressure relief spring **848**, upper end of lower stem **840** and ceramic control disk **844** are sized so that when the fluid pressure in the valve exceeds the predetermined pressure, lower stem lower end **842** raises off of ceramic base **854** by an amount that restricts the flow of fluid through outlet **860** to an amount that is between 50% and about 80% less than the normal flow of fluid through the valve when the valve is in its fully open position, or by a distance that allows between about 0.25 and about 1.25 GPM and about 0.10 and about 0.50 GPM to flow through the valve. Even more preferably, cartridge **800** is configured and the lengths of piston **846**, cavity **838**, pressure relief spring **848**, upper end of lower stem **840** and ceramic control disk **844** are sized so that when the fluid pressure exceeds the predetermined pressure, lower stem lower end **842** raises off of ceramic cartridge base **854** by a distance that allows an amount of fluid to flow from cartridge inlet **850** to cartridge outlet **860** that is less than about 0.40 GPM. Additionally, the length of lower stem **830** can be sized to define a larger or small ceramic disk gap **866**, which in turn also allows or restricts more or less of the fluid flow through low-pressure outlet **860**.

[0103] FIG. 12A shows cartridge **800** in its closed position and under normal fluid pressure conditions. Under these conditions, as FIG. 12A shows, piston **846** is in its fully extended position whereby the force of spring **848** forces piston **846** down, which in turn forces lower stem **830** down, until ceramic control disk **844** rests on ceramic cartridge base **854** and prevents fluid from flowing between ceramic cartridge base **854** and ceramic control disk **844** and out of outlet **860**.

[0104] FIG. 12B shows cartridge **800** under pressurized fluid conditions. Under these conditions, as FIG. 12B shows, piston **846** is in a retracted position whereby the force of the pressurized fluid overcomes the force of spring **848**, compressing it and forcing piston **846** up, which in turn forces

lower stem **830** up thereby defining ceramic disk gap **866** between ceramic cartridge base **854** and ceramic control disk **844**, at least a portion of ceramic disk gap **866** aligns with low-pressure outlet **860** such that an amount of fluid flows out outlet **860** when the spindle **836** has positioned ceramic control disk **844** so that its opening does not align with opening **860** and under normal conditions fluid would not flow through the valve.

[0105] Cartridge **800** can be used in single-spindle ceramic control valve faucets. Cartridge **800** is used by inserting its valve end **802** into a control valve of faucet housing, such as a faucet like that shown in FIGS. 17, 26 and 28 of U.S. Pat. No. 10,260,642. Cartridge **800** may be held in faucet housing by any means and need not have exterior O-ring **814** holding the cartridge **800** in faucet housing but could have threads or be held therein by other conventional means.

[0106] FIG. 13 is a perspective view of lower stem **830** of cartridge **800** according to the disclosure. Piston **846** is fixed to lower stem upper end **840** and configured to rotate with upper stem **822** when spindle **836** is rotated to open or close the valve yet slide longitudinally relative to upper stem **822** so that it may be inserted into upper stem **822** and extend and retract under the force of spring **848** and pressurized fluid. Specifically, as shown in FIG. 13, piston **846** has an elongate protrusion or key on the exterior of piston **868** extending radially outward toward and longitudinally down at least a portion of the length of piston **846**. Upper stem cavity **838** has a complementary recess in it that is configured to longitudinally, slidably receive the piston key **868** and hold it so that lower stem **830** rotates when upper stem **822** is rotated when spindle **836** is turned to open and close the valve by rotating ceramic control disk **844** to align or misalign its opening **858** with opening **860** and allow or prevent fluid from flowing from inlet **850** through cartridge base **806**, ceramic cartridge base **854**, opening **858** of ceramic control disk **844** to outlet **860**.

[0107] Piston **752** and **846** need not be solid cylinders and upper stems **722** and **822** need not have cylindrical cavities **742** and **838**. Rather, pistons **752** and **846** could be tubular in shape and stems **722** and **822** could have solid cores which slide into tubularly shaped pistons and have cylindrical slots in stems **722** and **822** shaped to receive them. Alternatively, pistons **752** and **846** could be rectangular in shape with complementarily shaped cavities **724** and **838**. Further, multiple pistons and springs could be used with complementary cavities. Even further, upper stems **722** and **822** could have pistons with lower stems **728** and **830** having complementary cavities. Additionally, multiple pistons and cavities could be used, if desired, for greater alignment and control.

Pipe-Burst-Prevention Washerless Cartridge

[0108] FIG. 14 depicts another embodiment of a valve system according to the disclosure. A single-handle washerless valve system **900** is shown having a cartridge **902** having a control stem **904** operably connected to a cartridge body **906** having hot and cold high pressure fluid supply openings **908** and **910** and a low-pressure fluid delivery opening **912** and over which a cap **914** and handle **916** are disposed, the handle **916** being held in place by a set screw **918** which is screwed into a hole in control stem **904**. Other means could be used to attach handle **916**. The cartridge openings **908**, **910** and **912** correspond to openings **920**, **922**

(not shown) and 924 in faucet 926, and seals 928, 930 and 932, respectively. Cartridge body 906 is disposed in recess 934 within faucet 926 with openings 908, 910, and 912 aligned with openings 920, 922 (not shown), and 924, with seals 928, 930, and 932, respectively, held therebetween. Calibrated spring 936 is disposed on cartridge body 906 and held in place by cap 914 screwed into hole in stem 904. Thus, set screw 918 screwed into handle 916 and control stem 904 with cap 914 on spring 936 which in turn is disposed on cartridge 906 with washers 932, 928 and 930 therebetween in recess 934 on faucet 926 aligned as described not only holds handle 916 in place but also cap 914 with spring 936 on cartridge 906 and washers 932, 928 and 930 therebetween in recess 934 on faucet 926.

[0109] Handle 916 is rotated right or left to adjust the temperature of the fluid flowing out of faucet 926 which aligns or misaligns openings 908 and 910 with 920 and 922 to allow more or less of hot or cold fluid to flow, thereby controlling the amount that hot 920 or cold 922 fluid supplied and the temperature of the fluid flowing out of outlet 924. Handle 916 is lifted or lowered to control the amount of fluid flowing out of valve system 900 causing it to raise cartridge body 906 up proportionately to the distance handle 916 is raised off of seals 928, 930 and 932, and allowing water to flow through high-pressure faucet openings 920, 922 and out delivery opening 924.

[0110] Cartridge 902 is configured and the lengths of control stem 904, cartridge body 906, cap 914 and spring 936 are sized to allow cartridge body 906 to raise off faucet 926 when the pressure of the fluid in the faucet 926 is greater than the predetermined pressure of the spring 936 so that spring 936 is compressed and an amount of fluid flows through openings 908, 910, 912, 920, 922 and 924. Spring 936 applies a force on cartridge body 906 and in turn seals 928, 930 and 932 on openings 920, 922 and 924 to keep valve 900 closed under normal fluid pressure conditions. When the fluid pressure rises above a predetermined pressure which is the compressive force required to compress spring 936, the fluid pressure forces cartridge body 906 upward allowing fluid to flow from high-pressure faucet inlet openings 920 and 922 to low-pressure outlet opening 924.

[0111] Preferably, cartridge 902 is configured and the lengths of control stem 904, cartridge body 906, cap 914 and spring 936 are sized to allow cartridge body 906 to raise off faucet openings 920, 922 and 924 when the pressure of the fluid is greater than the predetermined pressure of the spring 936 to allow at an amount of fluid to flow from faucet inlets 920 and 922 to faucet outlet 924 that is less than the amount of fluid that would flow through outlet 924 when handle 916 is in its fully open position. More preferably, cartridge 902 is configured and the lengths of control stem 904, cartridge 906, cap 914 and spring 936 are sized to allow cartridge 902 to raise off faucet 926 when the fluid pressure exceeds the predetermined pressure, and allow an amount of fluid to flow through the faucet 926 by an amount that is between about 50% and about 80% less than the normal flow of fluid through the valve system 900 when the valve is in its fully open position, or by a distance that allows between about 0.25 and about 1.25 GPM and about 0.10 and about 0.50 GPM to flow through the valve. Even more preferably, cartridge 902 is configured and the lengths of control stem 904, cartridge body 906, cap 914 and spring 936 are sized to allow cartridge body 906 to raise off faucet 926, when

handle 916 is in its closed position and the fluid pressure in faucet 926 exceeds the predetermined pressure, by a distance that allows an amount of fluid to flow through faucet openings 920, 922 and 924 that is less than about 0.40 GPM.

[0112] Spring 936 has a compressive force that allows valve system 900 to relieve pressure in the building plumbing system at the desired predetermined pressure. Thus, spring 936 is capable of compressing when the pressure of the fluid in cartridge 902 exceeds the predetermined pressure. Spring 936 has a predetermined activation pressure as described above according to the type of pipes and components in the building plumbing system.

[0113] Relief valves 100, 200 and 300, valve system 400, 500, 600 and 900, and cartridge 700, 800, and 902 may be made of using any material used to make conventional valves, valve systems and cartridges, including metals and metal alloys (e.g., brass, nickel, titanium, or stainless steel), polymeric or hard rubber materials (e.g., polystyrene-butadiene-styrenes ("SBS"), polyacrylonitrile-butadiene-styrenes ("ABS")), and any material disclosed in U.S. Pat. No. 10,260,642, which is hereby incorporated by reference in its entirety, and by any method used to make conventional valves, valve systems and cartridges, including those described in the '642 patent or U.S. Pat. Nos. 7,475,703, and 3,810,602, which are also hereby incorporated by reference.

Method of Preventing Pipe Bursts in a Building Plumbing System

[0114] A method of preventing pipe bursts due to freezing comprises the steps of disposing a pressure relief valve according to the disclosure, such as relief valves 100, 200 or 300, in fluid communication with a building plumbing system. Preferably, relief valve 100, 200 or 300 is disposed between a fluid source and a control valve or other fixture in the pipe for delivery water to an end user. More preferably, the method comprises disposing relief valve 100, 200 or 300 behind the same wall in which a control valve or other fixture in the building plumbing system is disposed.

[0115] A further method for preventing pipe bursts due to freezing can comprise disposing a pipe burst prevention valve system in a building plumbing system comprising a faucet housing having handle and outlet, control valve having inlet, and bypass having a relief valve according to the disclosure, such as relief valve 100, 200 or 300, disposed in fluid communication between one inlet and one outlet. Another method for preventing pipe bursts due to freezing can comprise disposing a pipe burst prevention valve system in a building plumbing system comprising a faucet housing having a handle and an outlet, a control valve having two inlets and a mixing valve cartridge, and a bypass having a relief valve according to the disclosure, such as relief valve 100, 200 or 300, disposed in fluid communication between one inlet and the outlet.

[0116] A method of preventing pipe bursts due to freezing in a building plumbing system further comprises the steps of disposing a fluid control valve system according to the disclosure in a building plumbing system. Fluid control valve system can be faucet valve system shown in FIGS. 17, 26 and 28 of U.S. Pat. No. 10,260,642, or FIG. 4, 6, or 8 herein with cartridge 700 and 800, or FIG. 14 with valve system 900 and cartridge 902. More specifically, a method for preventing pipe bursts due to freezing can comprise disposing a pipe burst prevention valve system in a building plumbing system that has a faucet housing having handle

and outlet, control valve **406A**, **406B**, **502A**, **502B**, or **900** according to the disclosure having an inlet, and a cartridge **700**, **800** or **902** according to the disclosure disposed in fluid communication between one inlet and one outlet.

[0117] “Fluid” and “water” as used herein are understood to be any liquid that can freeze. For a domestic plumbing system, the fluid typically is 100% pure water with associated minerals and salt and other additives or naturally occurring substances. However, fluid or water as used herein could be any liquid that has the potential to freeze in a building plumbing system when the system reaches the temperature at which the fluid freezes, such as any aqueous liquid having a sufficient amount of water in it to freeze under the building conditions.

[0118] It must be understood that despite the use of the singular “a” or “an” in a claim they cover more than one claimed element. For example, a claim to the relief valve according to the disclosure covers use of more than one relief valve according to the disclosure or a relief valve according to the disclosure and a cartridge or other fixture in a building plumbing system. Similarly, by way of another example, a claim to a cartridge according to the disclosure covers use of more than one cartridge according to the disclosure and a cartridge and a relief valve according to the disclosure.

[0119] It also will be understood that the claims are not limited to the embodiments described herein. The features of the pressure relief valve, cartridge, control valve system, and methods described are exemplary and may be altered, omitted, or adapted according to the requirements of the plumbing system function and features desired. It will also be understood that the embodiments described include the description herein as well as combinations and sub-combinations of any of the features and equivalents thereof, and any such combinations set forth in the original claims or any amendments thereto.

What is claimed is:

1. A relief valve for pipes or valves or other fixtures in a building plumbing system, comprising:

- a body having an inlet, an outlet, an exterior surface, and an interior surface that defines a generally longitudinal passageway that extends from the inlet to the outlet and is capable of allowing fluid to flow from inlet to outlet;
- a valve seat disposed on the interior surface of said body between the inlet and the outlet;
- a flow limiter disposed on the interior surface of said body between said valve seat and the outlet, and extending into the passageway; and
- a valve seat sealing portion disposed in fluid communication with said valve seat between the inlet and said flow limiter, and configured to seal said valve seat against the flow of fluid in the passageway when the fluid has a pressure that is less than a predetermined pressure.

2. The relief valve according to claim 1, wherein the predetermined pressure is less than about 1000 pounds per square inch.

3. The relief valve according to claim 2, wherein the predetermined pressure is less than about 500 pounds per square inch.

4. The relief valve according to claim 3, wherein the predetermined pressure is between about 200 and about 395 pounds per square inch.

5. The relief valve according to claim 1, wherein said flow limiter is configured to limit the flow of fluid through it to less than about 0.4 gallons per minute at a fluid pressure.

6. The relief valve according to claim 5, wherein said flow limiter is configured to limit the flow of fluid through it to from about 0.25 gallons per minute to about 1.25 gallons per minute.

7. The relief valve according to claim 5, wherein said flow limiter is comprised of a plurality of orifices.

8. The relief valve according to claim 1, wherein the valve seat is disposed generally parallel to the exterior surface of said body.

9. The relief valve according to claim 1, wherein the valve seat is disposed generally transverse to the exterior surface of said body.

10. The relief valve according to claim 1, wherein the valve seat sealing portion comprises a burst disk.

11. The relief valve according to claim 10, wherein the burst disk has a score.

12. A compact, pipe-burst-prevention valve system for use in a building plumbing system, comprising:

- a faucet housing having a handle and an outlet for delivering fluid to an end user;

- a control valve having an inlet and an outlet in fluid communication with each other and the control valve outlet in fluid communication with the faucet outlet; and

- a relief valve, comprising:

- a body having an inlet in fluid communication with the control valve inlet, an outlet in fluid communication with the control valve outlet, an exterior surface, and an interior surface that defines a generally longitudinal passageway that extends from the relief valve body inlet to the relief valve body outlet;

- a valve seat disposed on the interior surface of said body between the relief valve body inlet and outlet;

- a flow limiter disposed on the interior surface of said body between said valve seat and the relief valve body outlet and extending into the passageway; and

- a valve seat sealing portion disposed in fluid communication with said valve seat between the relief valve body inlet and said flow limiter, and configured to seal said valve seat against the flow of fluid when the fluid has a pressure that is less than a predetermined pressure.

13. The valve system according to claim 12, wherein said a control valve has a housing and said relief valve is disposed in the control valve housing.

14. The valve system according to claim 12, wherein the relief valve flow limiter is comprised of a plurality of orifices.

15. The valve system according to claim 12, wherein the valve seat is disposed generally parallel to the exterior surface of said relief valve body.

16. The relief valve according to claim 12, wherein the valve seat is disposed generally transverse to the exterior surface of said relief valve body.

17. The valve system according to claim 12, further comprising:

- a second control valve having an inlet and an outlet in fluid communication with each other and the second control valve outlet in fluid communication with the faucet outlet.

18. A method of preventing pipes from bursting, comprising the steps of:

disposing a pressure relief valve in fluid communication with a building plumbing system, the pressure relief valve, comprising:

a body having an inlet, an outlet, an exterior surface, and an interior surface that defines a generally longitudinal passageway that extends from the inlet to the outlet and is capable of allowing fluid to flow from inlet to outlet;

a valve seat disposed on the interior surface of said body between the inlet and the outlet;

a flow limiter disposed on the interior surface of said body between said valve seat and the outlet, and extending into the passageway; and

a valve seat sealing portion disposed in fluid communication with said valve seat between the inlet and said flow limiter, and configured to seal said valve seat against the flow of fluid in the passageway when the fluid has a pressure that is less than a predetermined pressure.

19. The method according to claim **18**, wherein the pressure relief valve is disposed in fluid communication with the building plumbing system between a fluid source and a fixture for delivering water to an end user.

20. The method according to claim **18**, wherein the relief valve flow limiter is comprised of a plurality of orifices.

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